Technische Universität Dresden

Fakultät Mathematik

Discrete Optimization

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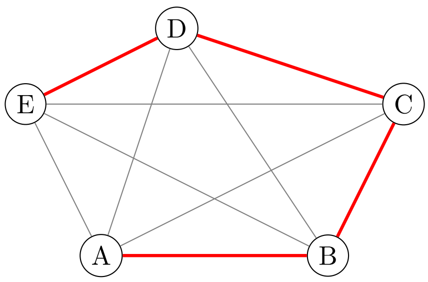

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1 Introduction

1.1 Motivation

Knapsack problem:
Weight limit: 12kg.
Things to put into the knapsack: $U_1, \dots, U_n$ with weights $a_1, \dots, a_n$ (in kg), utility value of $U_i : c_i$
Goal: Find a vector $x^* = (x_1^*, \dots, x_n^*)^T \in \{0,1\}^n$ (binary vector) so that the number of all utilities in the knapsack become maximal, whereas the weight of the things in the knapsack is limited by 12kg.

$$c^T x^* \geq c^T x \quad \text{for all } x \in \{\{0,1\}^n \mid a^T x \leq 12\},\ a^T x^* \leq 12$$

General knapsack problem: Maximize $c^T x$ subject to $a^T x - a_0 \leq 0$, $x \in D$, $a \in \mathbb{R}_+^n$, $a_0 > 0$, $c_i > 0$ for $i \in \{1, \dots, n\}$. If $D = \mathbb{R}_+^n$, then the knapsack problem is called *continuous*. If $D = \{0,1\}^n$ we have a *binary* knapsack problem. For $D = \mathbb{Z}^n$ we have an *integer* knapsack problem.

Facility Locations:

- Customers $k \in K := \{1, \dots, m\}$ need services from a company.
- $S := \{1, \dots, n\}$ set of possible locations.

Goal: Make a reasonable choice of locations.
The costs to serve a customer $k \in K$ from location $s \in S$ are given by $d_{ks} \geq 0$. Opening a new shop s costs $c_s \in \mathbb{R}$ (once).
Mathematical model:

$$\sum_{s \in S} c_s x_s + \sum_{k \in K} \sum_{s \in S} d_{ks} y_{ks} \quad \text{such that} \quad \begin{cases} \sum_{s \in S} y_{ks} = 1 & \forall k \in K \\ y_{ks} \leq x_s & \forall k \in K, s \in S \\ y_{ks} \geq 0 & \forall k \in K, s \in S \\ x_s \in \{0,1\} & \forall s \in S \end{cases}$$

The problem is known as *uncapacitated facility location* (UFL) problem. This problem is a *mixed-integer problem* (MIP) as x_s is an integer variable and y_{ks} is a continuous variable.

Traveling Salesman Problem (TSP):
Cities $S_1, \dots, S_n$. A person starts at S_1 and visits all other cities just once and, at the end of the travel, comes back to S_1.

$$E := \{(i,j) : \text{there is a direct way from city } S_i \text{ to city } S_j\}$$

if (i,j) belongs to E, then (j,i) need not belong to E. $c_{ij} \in \mathbb{R}$ are given for all $(i,j) \in E$.
insert image lecture notes.
Here $n = 5$. Therefore $E_1 := \{(1,2), (2,3), (3,1), (4,5), (5,4)\} \subseteq E \subseteq \{1,2,3,4,5\}^2$ and
$x_{ij} = \begin{cases} 1, & \forall (i,j) \in E_1, \\ 0, & \forall (i,j) \in E \setminus E_1. \end{cases}$

$$\sum_{s \in S} c_s x_s + \sum_{k \in K} \sum_{s \in S} d_{ks} y_{ks} \to \min \quad \text{such that } \begin{cases} \dots\dots \end{cases}$$

Minimize $\displaystyle\sum_{(i,j)\in E:\ i\in U,\ j\in V\setminus U} c_{ij}x_{ij}$, where $V=\{1,\dots,n\}$ such that

$$
\begin{aligned}
&\sum_{i:\ (i,j)\in E} x_{ij} = 1 \quad \text{for all } j\in\{1,\dots,n\},\\
&\sum_{j:\ (i,j)\in E} x_{ij} = 1 \quad \text{for all } i\in\{1,\dots,n\},\\
&\sum_{(i,j)\in E: i\in U, j\in V\setminus U} x_{ij} \geq 1 \quad \text{for all } U\in\mathcal{U}.\\
&x\in\{0,1\}^{|E|}
\end{aligned}
\tag{TSP}
$$

The first two conditions ensure we enter and leave every city just once. The third condition excludes subtours. We set $\mathcal{U} := \{U\subset V : Z\leq |U|\leq |V|-Z\}$.

Quadratic Assignment Problem (QAP):

- n persons
- n offices
- Person i needs to meet person j c_{ij} times per day ($c_{ij}\geq 0$).
- office k and office l are seperated by the distance $d_{kl}\geq 0$.
- we are looking for a map $\pi:\{1,\dots,n\}\to\{1,\dots,n\}$ which assigns each person i the office j so that the total walking distance becomes minimal.
- The mapping π is bijective (each person has one office and each office belongs to one person). Also $\pi\in S_n$ (permutation group)

$$f(\pi) := \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}d_{\pi(i)}d_{\pi(j)}.$$

We are looking for a permutation $\pi^*\in S_n$ such that $f(\pi^*)\leq f(\pi)$ for all $\pi\in S_n$. This is the *Quadratic Assignment Problem (QAP)*. Define

$$x_{ik} := \begin{cases} 1, & \text{if person } i \text{ has office } k,\\ 0, & \text{otherwise.}\end{cases}$$

QAP formulation: minimize

$$\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n}\sum_{l=1}^{n} c_{ij}d_{kl}x_{ik}x_{jl}$$

such that

$$\begin{cases} \displaystyle\sum_{k=1}^{n} x_{ik} = 1 \quad \text{for all } i\in\{1,\dots,n\}\\ \displaystyle\sum_{i=1}^{n} x_{ik} = 1 \quad \text{for all } k\in\{1,\dots,n\}\\ x_{ik}\in\{0,1\} \quad \text{for all } i,k\in\{1,\dots,n\}\end{cases}$$

1.2 Basic formulation of optimization problems and notions

Let $G \subseteq \mathbb{R}^n$ and $f : G \to \mathbb{R}$ be given. Then, we can consider the following optimization problem numbers need to be changed

$$\min_{x \in G} f(x) \tag{1}$$

f is called *objective function*, G is called *feasible set*. If $G = \mathbb{R}^n$ then (1) is called *free optimization problem*, otherwise, if $G \subsetneq \mathbb{R}^n$, then (1) is called *optimization problem*. Problem (1) means that we want to determine some $x^* \in G$ such that

$$f(x^*) \leq f(x) \quad \text{for all } x \in G. \tag{2}$$

x^* is then called *solution* of (1). The function value of $f(x^*)$ is called *optimal value* of (1). The existence of an optimal solution is not for sure.
Discrete feasible sets are often described by the means of $\mathbb{Z}^n$ (integer vectors of dimension n) or $\mathbb{B}^n = \{0, 1\}^n$. For example, the feasible set of a discrete optimization problem can be given by

$$G := \{x \in \mathbb{Z}^n : g_i(x) \leq 0 \text{ for } i \in I, h_j(x) = 0 \text{ for } j \in J\},$$

where I, J are finite index sets.
Set $G := \{x \in \mathbb{Z}^n : Ax = a, Bx \leq b\}$. Integer linear optimization problem

$$\min_{x \in G} c^T x.$$

Quadratic functions $x^T Cx + c^T x$. Integer quadratic optimization problem

$$\min_{x \in G} x^T Cx + c^T x.$$

2 Linear Optimization

2.1 Basics

For more details, see script in OPAL. 2.1 is not relevant for the exam!

$$\min c^T x \quad \text{s.t. } Ax = b, x \geq 0 \tag{LP}$$

in standard form.

2.2 Cutting optimization

Modeling of a 1-dimensional cutting problem. Length of the tube is $L > 0$. Different types of smaller tubes $T_1, \ldots, T_m$ of lengths $l_\nu \leq L$ ($\nu = 1, \ldots, m$), production needs b_ν ($\nu = 1, \ldots, m$), $b_\nu \in \mathbb{N}$. There exists only a finite number of variants to cut the long tube. For each variant, we define a variable $z_i \in \mathbb{Z}_+$ ($i = 1, \ldots, n$). z_i tells you how many long tubes are cut according to variant i.
Integer linear optimization problem:

$$\text{minimize} \sum_{i=1}^{n} z_i \quad \text{s.t. } Hz \geq b, z \geq 0, z \in \mathbb{Z}^n. \tag{27}$$

Column h_i of $H \in \mathbb{Z}^{m \times n}$ belongs to the i-th variant of the cutting of the large tube, the elements in h_i show how often the smaller tube T_ν is cut if variant i is used.

2.3 Polyhedra

Definition 2.1.
Let $C \in \mathbb{R}^{m \times n}$ be given. The rows of C are denoted by $c_1^T, \ldots, c_m^T$. Then

$$P(C, d) := \{x \in \mathbb{R}^n \mid Cx \leq d\} = \{x \in \mathbb{R}^n \mid c_i^T x \leq d_i \ \forall i \in I\}$$

is called *polyhedron* in $\mathbb{R}^n$, where $I := \{1, \ldots, m\}$. A bounded polyhedron is also called *polytope*. A polyhedron is said to be a *polyhedral cone*, if $d = 0$. A polyhedron P is called *rational*, if there exist $C' \in \mathbb{Q}^{m \times n}$ and $d \in \mathbb{Q}^m$ so that $P = P(C', d')$.

In the following, if nothing else is said, we assume (implicitly) that any polyhedron $P(C, d)$ is rational and also the data C', d are rational.

Definition 2.2.
Let $X, Y \subseteq \mathbb{R}^n$ be nonempty sets and $\mathcal{B} := \{x \in \mathbb{R}^n \mid ||x|| \leq 1\}$ denote the unit ball in $\mathbb{R}^n$, where $|| \cdot ||$ is any arbitrary norm in $\mathbb{R}^n$. Then the sets

$$X - Y := \{x - y \mid x \in X, y \in Y\} \quad \text{and} \quad X + Y := \{x + y \mid x \in X, y \in Y\}$$

are defined. Further

$$\text{aff}(X) := X + \text{lin}(X - X)$$

is called *affine hull* of X, where $\text{lin}(X) := \left\{ \sum_{i=1}^{k} \alpha_i x^i \mid x^i \in X, \alpha_i \in \mathbb{R}, k \in \mathbb{N} \right\}$. Further,

$$\operatorname{conv}(X) := \left\{ \sum_{i=1}^{t} \lambda_i x^i \mid x^1, \dots, x^t \in X,\ \sum_{i=1}^{t} \lambda_i = 1,\ \lambda_1, \dots, \lambda_t \geq 0, t \in \mathbb{N} \text{ with } 1 \leq t < \infty \right\}$$

is called *convex hull* of X and

$$\operatorname{cone}(X) := \left\{ \sum_{i=1}^{t} \lambda_i x^i \mid x^1, \dots, x^t \in X,\ \lambda_1, \dots, \lambda_t \geq 0, t \in \mathbb{N} \text{ with } 1 \leq t < \infty \right\}$$

is said to be the *(convex) cone generated* from X. $\operatorname{rint}(X) := \{x \in X \mid \exists \varepsilon > 0 : (x + \varepsilon \mathcal{B}) \cap \operatorname{aff}(X) \subseteq X\}$ denotes the *relative interior* of X.

$$\operatorname{rbd}(X) := \operatorname{cl}(X) \setminus \operatorname{rint}(X)$$

is called *relative boundary* of X. The dimension of a set $X \subseteq \mathbb{R}^n$ is defined as

$$\dim(X) := \dim(\operatorname{lin}(X - X)), \quad \dim(\emptyset) = -1$$

Remark 2.3.
One can show that $\operatorname{aff}(X) = x + \operatorname{lin}(X - X)$ for any $x \in X$. Further, one can realize that $\operatorname{aff}(X)$ (or $\operatorname{conv}(X)$) is the smallest affine (convex) set in $\mathbb{R}^n$ which contains X. $\operatorname{cl}(X)$ means the closure of X. It can be shown that a polyhedron $P(C, d) = P$ in $\mathbb{R}^n$ has exactly dimension n (full dimension) if P has an interior point, i.e., $\operatorname{int}(P) \neq \emptyset$. If there exist $\bar{x} \in \mathbb{R}^n$ with $(C\bar{x})_i < d_i$ for $i = 1, \dots, m$, then $P(C, d)$ has an interior point.

Definition 2.4.
Let P be a polyhedron in $\mathbb{R}^n$, $p \subseteq \mathbb{R}^n$, and $p_0 \in \mathbb{R}$ be given. Then the inequality

$$p^T x - p_0 \leq 0$$

($[p, p_0]$ for short) is called *valid inequality* for P if this inequality is satisfied for all $x \in P$

Definition 2.5.
Let P be a given polyhedron in $\mathbb{R}^n$. Moreover, let $[p, p_0]$ be a valid inequality for P and

$$F := \{x \in P \mid p^T x - p_0 = 0\}.$$

Then F is called *face* of P and one says that $[p, p_0]$ represents the face F. A face of P is called *proper* if $F \neq \emptyset$ and $F \neq P$.

Abbreviation: $[p, p_0]_= := \big\{x \in \mathbb{R}^n \mid p^T x - p_0 = 0\big\}$.
With this, we can write $F = P \cap [p, p_0]_=$ for a face F that is represented by $[p, p_0]$.

Lemma 2.6.
Let P be a polyhedron and $[p, p_0]$ be a valid inequality for P. Then $[p, p_0]$ represents

a nonempty face F of P if and only if

$$\max\{p^T x \mid x \in P\} = p_0.$$

Proof. No proof needed. □

Lemma 2.7.
Let $P := P(C,d) \subset \mathbb{R}^n$ be a nonempty polyhedron with $\operatorname{rank}(C) = k$ and let F be a nonempty face of P. Then, the following assertions hold:

(a)
$$F = R := \{x \in \mathbb{R}^n \mid c_i^T x = d_i \ \forall i \in I_=(F),\ c_i^T x \le d_i \ \forall i \in I_<(F)\}$$
with $I_=(F) := \{i \in I \mid c_i^T x = d_i \ \forall x \in F\}$, and $I_<(F) := I \setminus I_=(F)$.

(b) $\dim(F) \ge n - k$.

(c) P has a face of dimension $n - k$.

Proof. (a) "$x \in F \implies x \in R$": Let $x \in F$. From the definition of $I_=(F)$, we have $c_i^T x = d_i$, for all $i \in I_=(F)$. Since, for any face F of P, $F \subseteq P$, it follows $c_i^T x \le d_i$ for all $i \in I$ and especially for $i \in I_<(F)$. Thus, $x \in R$.

"$x \in R \implies x \in F$": The definition of R implies that $R \subseteq P$. Therefore, in order to show $R \subseteq F$, it suffices to show that $(P \setminus F) \cap R = \emptyset$. We would like to show that for any $y \in P \setminus F$ an index $j \in I_=(F)$ exists with $c_j^T y < d_j$. To this end, let $y \in P \setminus F$ be arbitrarily chosen. For each $j \in I_<(F)$ there is by definition a

$$\widehat{x}_j \in F \text{ with } c_j^T \widehat{x}_j < d_j. \tag{29}$$

Since F is convex, we have

$$\widehat{x} := \frac{1}{|I_<(F)|} \sum_{j \in I_<(F)} \widehat{x}_j \in F$$

and it follows (with (29))

$$c_i^T \widehat{x} = \frac{1}{|I_<(F)|} \sum_{j \in I_<(F)} c_i^T \widehat{x}_j < \frac{1}{|I_<(F)|} \sum_{j \in I_<(F)} d_j < d_j \quad \forall i \in I_<(F)$$

At this point the profesor got confused. Actual proof in OPAL

(b) One can see that $\operatorname{lin}(F - F) \subseteq L := \{x \in \mathbb{R}^n \mid c_i^T x = 0 \ \forall i \in I_=(F)\}$. To see that $\operatorname{lin}(F - F) \supseteq L$ we first note that there is $\widehat{x} \in F$ with $c_i^T \widehat{x} < d_i$ for all $i \in I_<(F)$. For $y \in L$ there is some $\widehat{\lambda} \in \mathbb{R} \setminus \{0\}$ so that

$$\begin{aligned} c_i^T(\widehat{x} + \widehat{\lambda} y) &\le d_i \quad \forall i \in I_<(F) \\ c_i^T(\widehat{x} + \widehat{\lambda} y) &= d_i \quad \forall i \in I_=(F). \end{aligned}$$

Because of part a), we have $\widehat{x} + \widehat{\lambda} y \in F$ or equivalently $\widehat{\lambda} y \in F - \widehat{x}$. This implies $y \in \operatorname{lin}(F - F)$. Hence, $\operatorname{lin}(F - F) = L$. Taking into account that $\operatorname{rank}(C) = k$, we get

$$\dim(F) = \dim(\operatorname{lin}(F - F)) = \dim(L) \ge n - k.$$

(c) Let F be a nonempty face of P with minimal dimension. If $\dim(F) = 0$, then $k = n$ (because of (b)) and nothing has to be shown. Therefore, let $\dim(F) > 0$. In analogy to the proof of part a) there exists $\widehat{x} \in F$ with $c_i^T \widehat{x} < d_i$ for all $i \in I_<(F)$. By $\dim(F) > 0$ it follows the existence of some $y \in F$ with $y \neq \widehat{x}$. Let us now assume that there is $\widehat{\lambda} \in \mathbb{R}$ with

$$z(\widehat{\lambda}) := \widehat{x} + \widehat{\lambda}(y - \widehat{x}) \in \{x \in F \mid c_i^T x = d_i \text{ for at least one } i \in I_<(F)\}.$$

Let $j \in I_<(F)$ with $c_j^T z(\widehat{\lambda}) = d_j$. Then

$$F_j := \{x \in P \mid c_i^T x = d_i \; \forall i \in I_=(F) \bigcup \{j\}\}$$

a nonempty face of P with $\dim(F_j) < \dim(F)$. Since F is already of minimal dimension, we have a contradiction and our assumption is wrong. Therefore, we have

$$z(\lambda) \notin F \text{ or } c_i^T z(\lambda) \neq d_i \quad \forall (\lambda, i) \in \mathbb{R} \times I_<(F). \tag{33}$$

Since $z(\lambda) \in F$ for all $\lambda \in [0, 1]$ it follows

$$c_i^T z(\lambda) < d_i \quad \forall (\lambda, i) \in [0, 1] \times I_<(F). \tag{34}$$

Moreover, it holds that

$$c_i^T z(\lambda) = d_i \quad \forall (\lambda, i) \in \mathbb{R} \times I_=(F) \tag{35}$$

Let us assume that there is $\lambda \in \mathbb{R} \setminus [0, 1]$ with $z(\lambda) \notin F$. If $\lambda > 1$ ($\lambda < 0$), then we get by reducing (or increasing) λ a $\bar{\lambda} \in \mathbb{R} \setminus [0, 1]$ and $j \in I_<(F)$ by using (34) and (35) that $z(\bar{\lambda}) \in F$ and $c_j^T z(\bar{\lambda}) = d_j$. This contradicts (33). Therefore we, have $z(\lambda) \in F$ for all $\lambda \in \mathbb{R}$, i.e.

$$Cz(\lambda) = C\widehat{x} + \lambda C(y - \widehat{x}) \leq d \quad \forall \lambda \in \mathbb{R}. \tag{36}$$

This implies $C(y - \widehat{x}) = 0$ since otherwise a contradiction to (36) would follow. Since the arguments used by now are valid for every $y \in F \setminus \{\widehat{x}\}$, we have

$$F \subseteq \{y \in \mathbb{R}^n \mid Cy = C\widehat{x}\} = \widehat{x} + \{x \in \mathbb{R}^n \mid Cx = 0\}$$

and therefore, $\dim(F) \leq \dim(\ker(C)) = n - k$. Because of (b), we have $\dim(F) = n - k$.

□

Lemma 2.8.
Let F and F' be nonempty faces of a polyhedron P with $F \subseteq F'$ and $\dim(F) = \dim(F')$. Then it holds that $F = F'$.

Proof. From $F \subseteq F'$ it follows that $\text{lin}(F - F) \subseteq \text{lin}(F' - F')$. Since $\dim(F) = \dim(F')$, we get (taking into account Definition 2.2) that $\dim(\text{lin}(F - F)) = \dim(\text{lin}(F' - F'))$ and, therefore,

$$\text{lin}(F - F) = \text{lin}(F' - F'). \tag{37}$$

For $y' \in F'$ and $y \in F$, we have that $y - y' \in \operatorname{lin}(F' - F')$ (thanks to $F \subseteq F'$). Because of (37), there exist vectors $z_i \in F - F$ and numbers $\mu_i \in \mathbb{R}$ (with $i = 1, ..., k := \dim(F)$) so that

$$y - y' = \sum_{i=1}^{k} \mu_i z_i \quad \text{or equivalently} \quad y' = y - \sum_{i=1}^{k} \mu_i z_i$$

We now take an inequality $[p, p_0]$ which represents the face F. Then we have

$$p^T y' = p^T y - \sum_{i=1}^{k} \mu_i p^T z_i = p_0,$$

and, thus $y' \in \{x \in \mathbb{R} \mid p^T x - p_0 = 0\} = F$, i.e. $F' = F$ follows. □

Definition 2.9 (facet).
Let P be a polyhedron with $\dim(P) \geq 1$. A face F of P is called *facet of* P, if $\dim(F) = \dim(P) - 1$.

Theorem 2.10.
Let F be a facet of the polyhedron $P := P(C, d)$. Then there exists an index $i \in I_<(P) := \{i \in I \mid c_i^T x < d_i$ for at least one $x \in P\}$ so that $[c_i, d_i]$ represents the facet F.

Proof. We first see that $I_<(P) \neq \emptyset$ since, otherwise $P = \{x \in \mathbb{R}^n \mid c_i^T x = d_i \ \forall i \in I\}$ would be an affine subspace, which cannot possess a facet at all. We are now going to show that $P \cap [c_i, d_i]_= = F$ for at least one $i \in I_<(P)$. This is equivalent to saying $P \cap [c_i, d_i]_= = F \cap [c_i, d_i]_=$ for at least one $i \in I_<(P)$. To prove this, we assume the contrary, i.e., that $P \cap [c_i, d_i]_= \supsetneq F \cap [c_i, d_i]_=$ for all $i \in I_<(P)$.
As $i \in I_<(P)$, it holds that $P \supsetneq P \bigcap [c_i, d_i]$ and, with Lemma 2.8, we get $\dim (F \cap [c_i, d_i]_=) + 1 \leq \dim (P \cap [c_i, d_i]_=) = \dim(P) - 1$ for all $i \in I_<(P)$.
For this, we have taken into account Lemma 2.8. It says that the dimensions of nonempty faces F_1, F_2 of P with $F_1 \subsetneq F_2$ have a difference of at least 1. One should keep in mind as well that P, $P \cap [c_i, d_i]_=$, F and $F \cap [c_i, d_i]_=$ are faces of P, where only $F \cap [c_i, d_i]_=$ could be empty. From the previous inequality chain, we get $\dim (F \cap [c_i, d_i]_=) \leq \dim(P) - 2 \quad \forall i \in I_<(P)$. Since F is a facet of P, it holds that $\dim(F) = \dim(P) - 1$. Thus, $F \supsetneq F \cap [c_i, d_i]_= = F \cap \{x \in \mathbb{R}^n \mid c_i^T x - d_i = 0\}$ for all $i \in I_<(P)$. Therefore, for each $j \in I_<(P)$, there exists $x(j) \in F$ with

$$c_j^T x(j) < d_j \tag{38}$$

For any inequality $[p, p_0]$ representing F and any $j \in I_<(P)$, we therefore have

$$p^T x(j) - p_0 = 0 \tag{39}$$

Setting $\widehat{x} := \dfrac{1}{|I_<(P)|} \displaystyle\sum_{j \in I_<(P)} x(j)$, we obtain $\widehat{x} \in F$ (by convexity of F). and

$$c_i^T \widehat{x} < d_i \quad \forall i \in I_<(P). \tag{40}$$

(40) follows from (38) and $c_i^T x(j) \leq d_i$ for all $i \in I$ since $\widehat{x} \in F \subset P$. Since $\dim(F) = \dim(P) - 1$, there exists $\widehat{y} \in P \setminus F$. Thus $p^T \widehat{y} - p_0 < 0$ follows. Setting $z(\varepsilon) := \widehat{x} + \varepsilon(\widehat{x} - \widehat{y})$

for $\varepsilon \in \mathbb{R}$, we get, using $\widehat{x} \in F$ and $\widehat{y} \in P \setminus F$, that $z(\varepsilon) \in \operatorname{aff}(P)$. For $\varepsilon > 0$, we further obtain

$$p^T z(\varepsilon) = p^T \widehat{x} + \varepsilon p^T(\widehat{x} - \widehat{y}) = p_0 + \overbrace{\varepsilon(p_0 - p^T \widehat{y})}^{>0} > p_0,$$

i.e., $z(\varepsilon) \notin P \ \forall \varepsilon > 0$. Thus, $\widehat{x} \in \operatorname{rbd}(P)$. On the other hand, we know $\operatorname{aff}(P) = \widehat{x} + \operatorname{lin}(P - P)$. For any $q \in \operatorname{lin}(P - P)$, it holds $c_i^T q = 0$ for all $i \in I_=(P) := \{i \in I \mid c_i^T x = d_i \ \forall x \in P\} = I \setminus I_<(P)$. Hence, using (40), we obtain that, for any $q \in \operatorname{lin}(P - P)$ and any $\varepsilon > 0$ sufficiently small,

$$c_i^T(\widehat{x} + \varepsilon q) < d_i \quad \forall i \in I_<(P) \quad \text{and} \quad c_i^T(\widehat{x} + \varepsilon q) = d_i \quad \forall i \in I_=(P).$$

Thus, $\widehat{x} \in \operatorname{rint}(P)$ follows, which contradicts $\widehat{x} \in \operatorname{rbd}(P)$. Therefore, the assumption in the beginning of the proof is wrong and the assertion of the theorem is true. □

Corollary 2.11.
For the description of a polyhedron P as solution set of a system of linear inequalities, for each facet of P this system must contain an inequality that represents this facet.

Theorem 2.12.
Let $P := P(C, d)$ be a polyhedron. Further, $[c_j, d_j]$ with $j \in I_<(P)$ be an inequality which represents a face F of P with $\dim(F) < \dim(P) - 1$. Then, it holds that

$$P(C, d) = \{x \in \mathbb{R}^n \mid c_i^T x \leq d_i \quad \forall i \in I \setminus \{j\}\},$$

i.e., the inequality $[c_j, d_j]$ is redundant and can be removed from the description of the polyhedron.

Proof. Exercise. □

Theorem 2.13 (Minkowski-type Theorem).
A set $P \subset \mathbb{R}^n$ is a rational polyhedron if and only if there exist finite sets $S = \{x^1, ..., x^q\} \subset \mathbb{Q}^n$ and $T = \{y^1, ..., y^r\} \subset \mathbb{Q}^n$ so that

$$P = \operatorname{conv}(S) + \operatorname{cone}(T).$$

(an analogue of this theorem is known for real polyhedra (by Minkowski).)

Proof. No proof. □

2.4 Integer linear optimization

2.4.1 Basics

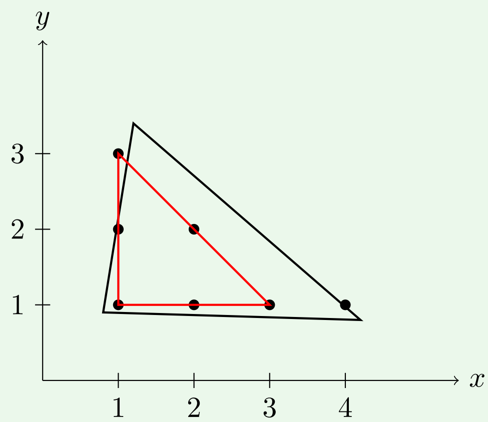
Let $c \in \mathbb{Q}^n$, $b \in \mathbb{Q}^m$, $A \in \mathbb{Q}^{m \times n}$ be given. Then we consider the optimization problem

$$\min c^T x \quad \text{such that } x \in G := Q \cap \mathbb{Z}^n, \tag{ILP}$$

where $Q := Q(A, b) := \{x \in \mathbb{R}^n \mid Ax \leq b, x \geq 0\}$.

$$\min_{x \in Q_I} c^T x \tag{CILP}$$

Definition 2.14 (Integer hull).
The set $Q_I := \operatorname{conv}(G) = \operatorname{conv}(Q \cap \mathbb{Z}^n)$ is called integer hull of Q. picture to be corrected



Lemma 2.15.
Q_I is a rational polyhedron.

Proof. No proof. □

Lemma 2.16.
Let $Q \cap \mathbb{Z}^n \neq \emptyset$. Then the following assertions hold:

(a) $\inf\{c^T x \mid x \in Q \cap \mathbb{Z}^n\} = -\infty \iff \inf\{c^T x \mid x \in Q_I\} = -\infty$.

(b) If $\inf\{c^T x \mid x \in Q_I\} > -\infty$, then (ILP) has at least one solution which is a vertex of Q_I. Every solution of (CILP) that is a vertex of Q_I solves (ILP).

(c) If $\inf\{c^T x \mid x \in Q \cap \mathbb{Z}^n\} > -\infty$, then (ILP) has a solution and every solution of (ILP) solves (CILP).

Proof. (a) " $\Longrightarrow$ ": Obviously, we have

$$-\infty = \inf\{c^T x \mid x \in Q \cap \mathbb{Z}^n\} \geq \inf\{c^T x \mid x \in Q_I\}$$

because $Q_I \supset Q \cap \mathbb{Z}^n$.
" $\Longleftarrow$ ": Due to Lemma 2.15, Q_I is a rational polyhedron. Therefore, Theorem 2.13 says that there are finite sets $S, T \subset \mathbb{Q}^n$ with $Q_I = \operatorname{conv}(S) + \operatorname{cone}(T)$. In particular, S and $\operatorname{conv}(S)$ are bounded and it holds

$$\sigma := \inf\{c^T x \mid x \in \operatorname{conv}(S)\} > -\infty.$$

Because of $\inf\{c^T x \mid x \in Q_I\} = -\infty$ there exists $x^0 \in Q_I$ with $c^T x^0 \leq \sigma - 1$. Hence, $x^0 \notin Q_I \setminus \operatorname{conv}(S)$ and there are $\tilde{x} \in \operatorname{conv}(S)$ and $\lambda_1, ..., \lambda_r \geq 0$ such that

$$x^0 = \tilde{x} + \sum_{i=1}^{r} \lambda_i y^i,$$

where $r \geq 0$ denotes the number of elements $y^i \in T$.
Then, it follows that $c^T x^0 = c^T \tilde{x} + \sum_{i=1}^r \lambda_i c^T y^i \leq \sigma - 1$ and further (due to $c^T \tilde{x} \geq \sigma$).

$$\sum_{i=1}^{r} \lambda_i c^T y^i \leq \sigma - 1 - c^T \tilde{x} \leq -1$$

Hence, there is $j \in \{1, ..., r\}$ such that $\lambda_j > 0$ and $c^T y^j \leq -\frac{1}{\lambda_j r}$. Because of $\emptyset \neq Q \cap \mathbb{Z}^n = Q_I \cap \mathbb{Z}^n$, there is at least one $\widehat{x} \in Q_I \cap \mathbb{Z}^n$. Taking into account that $Q_I = \operatorname{conv}(S) + \operatorname{cone}(T)$ and $y^j \in \mathbb{Q}^n$, we get

$$x(\nu) := \widehat{x} + \nu y^j \in Q \cap \mathbb{Z}^n$$

for infinitely many $\nu \in \mathbb{N}$ and

$$c^T x(\nu) = c^T \widehat{x} + \nu c^T y^j \leq c^T \widehat{x} - \frac{\nu}{\lambda_j r} \quad \forall \nu \in \mathbb{N}.$$

For $\nu \to \infty$ it follows $c^T x(\nu) \to -\infty$ and we thus have

$$\inf\{c^T x \mid x \in Q \cap \mathbb{Z}^n\} = -\infty.$$

(b) Exercise.

(c) Exercise. □

2.4.2 Integer polyhedra and totally unimodular matrices

Definition 2.17 (Integer polyhedron).
A nonempty polyhedron $P \subset \mathbb{R}^n$ is called integer polyhedron, if all nonempty faces of P contain an integer point.

Lemma 2.18.
Let $P := P(C, d) \subseteq \mathbb{R}^n$ be a nonempty polyhedron with $\operatorname{rank}(C) = n$. Then P is integer if and only if all vertices of P are integer.

Proof. " $\Longrightarrow$ ": Trivial.
" $\Longleftarrow$ ": Any nonempty face F of P is a polyhedron, since $F = \{x \in \mathbb{R}^n \mid Cx \leq d, p^T x = p_0\}$, where $[p, p_0]$ represents F. Consequently, the rank(C) with C extended by the rows p^T and $-p^T$ is equal to n. According to Lemma 2.7 c), F has a face of dimension 0, i.e., it has a vertex. Since we assumed that all vertices are integer, it follows that any nonempty face contains integer point. □

Corollary 2.19.
A nonempty polyhedron $P \subseteq \mathbb{R}^n_+$ is integer if and only if all vertices of P are integer.

Proof. For a description $P = P(C, d)$ one can choose C (due to $P \subseteq \mathbb{R}^n_+$) in such a way that it contains the (negative) identity matrix of dimension $n \times n$ and therefore $\operatorname{rank}(C) = n$ follows. Thus, using Lemma 2.18 completes the proof. □

Corollary 2.20.
Let $Q \cap \mathbb{Z}^n \neq \emptyset$. Then the polyhedron Q_I is integer.

Proof. Because $Q \cap \mathbb{Z}^n \neq \emptyset$ we also have that $Q_I \neq \emptyset$. Moreover, it is known that all vertices of Q_I are integer and $Q_I \subseteq Q \subseteq \mathbb{R}^n_+$. Thus, the result follows from Corollary 2.19. □

What are reasonable conditions for $Q_I = Q$? $Q = \{x \in \mathbb{R}^n \mid Ax \leq b, x \geq 0\}$

Definition 2.21.
A matrix $A \in \mathbb{Z}^{m \times n}$ is called <u>totally unimodular</u> (TU), if the determinant of any quadratic submatrix of A is 0, 1 or -1.

Let $\mathcal{I} \subseteq \{1, ..., m\}$ and $\mathcal{J} \subseteq \{1, ..., n\}$ be nonempty index sets with the same cardinality, i.e. $|\mathcal{I}| = |\mathcal{J}|$. Then the matrix $A_{\mathcal{I},\mathcal{J}} = (a_{ij})_{(i,j) \in \mathcal{I} \times \mathcal{J}}$ is a quadratic submatrix of A. Any quadratic submatrix of A can be described in this way.

Theorem 2.22.
Let $A \in \mathbb{Z}^{m \times n}$ be given. For $Q(b) := \{x \in \mathbb{R}^n \mid Ax \leq b, x \geq 0\}$, TFAE:

(a) $Q(b)$ is integer for all $b \in \mathbb{Z}^n$ with $Q(b) \neq \emptyset$.

(b) A is totally unimodular.

Proof. <u>(a) $\Longrightarrow$ (b):</u> Let A_{11} be an arbitrary $k \times k$-submatrix of A. WLOG, we may assume that $A_{11} = (a_{ij})_{1 \leq i,j \leq k}$. For $\det(A_{11}) = 0$, we are done. Therefore, we assume that A_{11} is nonsingular. In this case, we show that $|\det(A_{11})| = 1$. Let A be divided in submatrices as follows:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in \mathbb{Z}^{m \times n}.$$

Further, let

$$\tilde{A} := \begin{pmatrix} A_{11} & 0_{k,m-k} \\ A_{21} & I_{m-k} \end{pmatrix} \in \mathbb{Z}^{m \times m}.$$

Obviously, $\tilde{A}$ is nonsingular and it holds that $|\det(A_{11})| = |\det(\tilde{A})|$. To show $|\det(\tilde{A})| = 1$, we now prove that $\tilde{A}^{-1}$ (and therefore $\det(\tilde{A})$) is integer. Because

$$\underbrace{\det(\tilde{A})}_{\in \mathbb{Z}} \cdot \underbrace{\det(\tilde{A}^{-1})}_{\in \mathbb{Z}} = 1$$

it follows that $|\det(\tilde{A})| = 1$. Let $j \in \{1, ..., m\}$ and $e^j \in \mathbb{R}^m$ (j-th unit vector). Then there exists a vector $v \in \mathbb{Z}^m$ with $\bar{a} := v + \tilde{A}^{-1} e^j \geq 0$. Moreover, let $b := \tilde{A}\bar{a}$. Then, $\tilde{A}^{-1} b = \bar{a} \geq 0$. The vector

$$\bar{x} := \begin{pmatrix} \bar{a}_{1 \leq j \leq k} \\ 0_{n-k} \end{pmatrix} \in \mathbb{R}^n$$

is a vertex of $Q(b)$. This follows with Lemma 2.9 (german lecture notes, not in this version) from

$$\bar{x} \geq 0, \; A\bar{x} = \tilde{A}\bar{a} - \begin{pmatrix} 0_{k,m-k} \\ I_{n-k} \end{pmatrix} \bar{a}_{k+1 \leq i \leq m} = \begin{pmatrix} b_{1 \leq i \leq k} \\ (b - \bar{a})_{k+1 \leq i \leq m} \end{pmatrix} \leq b \tag{41}$$

i.e., $\bar{x} \in Q(b)$. Using the definition of $\bar{x}$ and (41), we get that $\bar{x}$ satisfies

$$x_{k+1\leq i\leq n} \geq 0,\ (Ax)_{1\leq i\leq k} \leq b_{1\leq i\leq k}$$

with equality, their gradients span (due to the nonsingularity of A_{11}) the $\mathbb{R}^n$. Therefore, by Lemma 2.9, $\bar{x}$ is a vertex of $Q(b)$ and $\bar{x} \in \mathbb{Z}^n$ (from assumption (a)).
Since $\tilde{A} \in \mathbb{Z}^{m\times n}$, $v \in \mathbb{Z}^m$ and $b \in \mathbb{Z}^m$, $A\bar{x} \in \mathbb{Z}^m$, $(b-\bar{a})_{k+1\leq i\leq m} \in \mathbb{Z}^{m-k}$, $\bar{a}_{1\leq i\leq k} \in \mathbb{Z}^k$. Therefore, $\bar{a}$ and $\tilde{A}^{-1}e^j = \bar{a} - v$ are integer vectors since $j \in \{1, ..., m\}$ is arbitrary, we get that $\tilde{A}^{-1} \in \mathbb{Z}^{m\times m}$.
<u>(b) $\implies$ (a):</u> Let A be TU, moreover let $b \in \mathbb{Z}^m$ be arbitrarily chosen so that $Q(b) \neq \emptyset$. According to Corollary 2.19 it is sufficient to show that every vertex of $Q(b)$ is integer. Let x' be a vertex of $Q(b)$. Then there exists (by Lemma 2.9) a nonsingular submatrix $C' \in \mathbb{Z}^{n\times n}$ of $C := \begin{pmatrix} A \\ -I \end{pmatrix} \in \mathbb{Z}^{(m+n)\times n}$ and a subvector $d' \in \mathbb{Z}^n$ of $d \in \mathbb{Z}^{m+n}$ such that $Cx' \leq d$ and $C'x' = d'$. Since A is a TU matrix, $\begin{pmatrix} A \\ -I \end{pmatrix}$ must be TU (see Lemma 2.23). Therefore, we must have $\det(C') = \pm 1$. With Cramer's rule it follows that $x' = (C')^{-1}d'$ is integer. □

Lemma 2.23.
Let $A \in \mathbb{Z}^{m\times n}$. TFAE:

(a) A is TU.

(b) A^T is TU.

(c) (A, I) is TU.

(d) A matrix which is the result of multiplying a row of A with -1 is TU.

(e) A matrix which is obtained from A by exchanging two rows of A is TU.

(f) A matrix which is obtained by adding the same row(s) is TU.

Proof. No proof given. □

Theorem 2.24.
Let $A \in \mathbb{R}^{m\times n}$. Let the following conditions be satisfied:

(1) Each column of A contains not more than two nonzero elements.

(2) Each entry of A is equal to 0, 1 or -1.

(3) The rows of A can be divided in two disjunct sets so that the following assertions hold

  (a) If in a column of A two nonzero elements appear with the same sign, then the two corresponding rows do not belong to the same set of rows.

  (b) If in a column of A two nonzero entries appear with a different sign, then the two corresponding rows belong to the same set of row.

Then A is totally unimodular.

Proof. No proof given. □

2.4.3 Valid inequalities

Inequalities which are valid for Q_I (but hopefully not for Q).
Chvatal-Gomory-rounding: $\lfloor \alpha \rfloor := \max\{z \in \mathbb{Z} \mid z \leq \alpha\}$ for all $\alpha \in \mathbb{R}$, $\lfloor a \rfloor := (\lfloor a_1 \rfloor, \ldots, \lfloor a_n \rfloor)^T$ for any $a \in \mathbb{R}^n$.

Theorem 2.25 (Chvatal-Gomory-rounding).
Let $[p, p_0]$ be a valid inequality for Q_I. Then $[\lfloor p \rfloor, \lfloor p_0 \rfloor]$ is a valid inequality for Q_I.

Proof. Since $x \geq 0$ for all $x \in Q_I$, we have

$$p^T x - \lfloor p \rfloor^T x = (p - \lfloor p \rfloor)^T x \geq 0 \quad \forall x \in Q_I.$$

Therefore, it follows that $\lfloor p \rfloor^T x \leq p^T x \leq p_0 \quad \forall x \in Q_I$. Thus, we have $\lfloor p \rfloor^T x - \lfloor p_0 \rfloor \leq 0$ for all $x \in Q_I \cap \mathbb{Z}^n$. Therefore, $\lfloor p \rfloor^T x - \lfloor p_0 \rfloor \leq 0$ holds for all $x \in Q_I$, because $Q_I = \mathrm{conv}(Q_I \cap \mathbb{Z}^n)$. $\square$

As valid inequality for Q_I (in Theorem 2.25) we can use (for example) $\left[\lfloor A^T u \rfloor, \lfloor b^T u \rfloor\right]$ with an arbitrary but fixed $u \in \mathbb{R}^m_+$. Then Theorem 2.25 yields the valid inequality for $Q_I : \left[\lfloor A^T u \rfloor, \lfloor b^T u \rfloor\right]$. An application of the Chvatal-Gomory-rounding is given for the case:

$$p^T x = p_0 \quad \forall x \in Q_I.$$

For example, if the description of Q contains linear equations, then $[p, p_0]$ and $[-p, -p_0]$ are valid inequalities for Q_I. Then $[p/l, p_0/l]$ for any $l \in \mathbb{N} \setminus \{0\}$ is a valid inequality for Q_I. The Chavatal-Gomory-rounding then yields $\left[\lfloor \frac{p}{l} \rfloor, \lfloor \frac{p_0}{l} \rfloor\right]$ or equivalently $\left[l\lfloor \frac{p}{l} \rfloor, l\lfloor \frac{p_0}{l} \rfloor\right]$.

2.4.4 The Branch and Bound method

We want to solve optimization problems of the form

$$\min_{x \in G} f(x), \tag{42}$$

where $G \subset \mathbb{R}^n$ is a discrete set and $f : G \to \mathbb{R}$. One way of solving (42) could be complete enumeration.

Algorithm 1: Branch & Bound

S1 (Start): $\mathcal{G} := \{G\}$, determine $y \in G$ and set $\widehat{b} := f(y)$.
S2 (Branch): CHOOSE $X \in \mathcal{G}$, $r \in \mathbb{N}$ with $r \geq 2$ and nonempty sets $X_1, ..., X_r$ so that

$$\bigcup_{1 \leq i \leq r} X_i = X \quad \text{and} \quad X_i \cap X_j = \emptyset \quad \forall i,j : i \neq j.$$

S3 (Bound): Determine $b(X_1), ..., b(X_r) \in \mathbb{R}$ (lower bounds for f on X_i) so that

$$\inf\{f(x) \mid x \in X_i\} \begin{cases} \geq b(X_i), & |X_i| > 1 \\ = b(X_i), & |X_i| = 1 \end{cases} \quad \text{for } i = 1, ..., r.$$

S4 (Update): $\mathcal{G} := \mathcal{G} \setminus \{X\}$ and $\mathcal{G} := \mathcal{G} \cup \{X_1, ..., X_r\}$
FORALL $Y \in \mathcal{G}$ DO
Determine $z \in Y$.
IF $f(z) < \widehat{b}$ THEN $\widehat{b} := f(z)$ and $y := z$
END FORALL
FORALL $Y \in \mathcal{G}$ DO
IF $b(Y) \geq \widehat{b}$ THEN $\mathcal{G} := \mathcal{G} \setminus \{Y\}$.
END FORALL
IF $\mathcal{G} \neq \emptyset$ THEN GOTO S2 ELSE $x^* := y$ and STOP.

Remark 2.26.
- The system $\mathcal{G}$ of sets contains always all subsets of the feasible set G, which have to be inspected, in the beginning $\mathcal{G}$ contains G.
- Computing good lower bounds $b(X_i)$ can be expensive but can be useful for the time the algrithm needs to stop.

Theorem 2.27.
Let the feasible set G of (42) satisfy $2 \leq |G| < \infty$. Then the branch and bound algorithm 1 is well defined and provides a solution of (42) after finitely many iterations (from step S2 to S4).

Proof. For self study. □

Remark 2.28.
- One may replace the condition $X_i \cap X_j = \emptyset$ by something weaker.
- There are different way to chose X from $\mathcal{G}$ in Step 2.
  1. **Best-bound search:** Among all sets $Y \in \mathcal{G}$, one chooses a set $X \in \mathcal{G}$ with the smallest lower bound, i.e., $b(X) = \min\{b(Y) \mid Y \in \mathcal{G}\}$.
  2. **Depth-first search:** (LIFO). Among all sets $Y \in \mathcal{G}$ with maximal number of predecessors, we choose a set with minimal lower bound.
  3. **Breadth-first search:** (FIFO) Among all sets $Y \in \mathcal{G}$ with minimal number of predecessors, we choose a set with minimal lower bound.
- Computing good upper bounds $\widehat{b}$ can improve the speed of the algorithm but may need time.

- Branch and Bound algorithm by Land/Doig and Dakin for (ILP)

$$\min_{x \in G = Q \cap \mathbb{Z}^n} c^T x$$

with a polyhedron Q. For description of the algorithm we will use $Q(X)$, where X denotes an element of $\mathcal{G}$. For the first iteration (in step S2) $X := G$ and $Q(G) := Q$. Every X chosen in a later iteration is described as intersection of the polyhedron $Q(X)$ and $\mathbb{Z}^n$.
Branch: A set $X = Q(X) \cap \mathbb{Z}^n$ chosen in step S2 is divided in two sets

$$X_1 := Q(X_1) \cap \mathbb{Z}^n \text{ with } Q(X_1) := \{x \in Q(X) \mid x_j \le \lfloor \widehat{x}_j \rfloor\}$$

$$X_2 := Q(X_2) \cap \mathbb{Z}^n \text{ with } Q(X_2) := \{x \in Q(X) \mid x_i \ge \lfloor \widehat{x}_j \rfloor + 1\}$$

$\widehat{x}$ is the solution of the linear program

$$\min_{x \in Q} c^T x$$

a solution is denoted by $\widehat{x}$. j denotes a component of $(\widehat{x}_1, ..., \widehat{x}_n)$ such that $\widehat{x}_j \notin \mathbb{Z}$.
Bound: For X_i, $(i = 1, 2)$, the optimal value of

$$\min_{x \in Q(X_i)} c^T x \tag{4.3}$$

is used as lower bound $b(X_i)$.

Solution of the TSP by branch & bound: Cost matrix $C = (c_{ij}) \in (\mathbb{R}_+ \cup \{\infty\})^{n \times n}$ with $c_{ij} = \infty$ for $i = 1, ..., n$.. Reduced cost matrix: $C^* = (c^*_{ij})$

FOR $i = 1, ..., n$ DO
  $\alpha_i := \min\{c_{ij} \mid j = 1, ..., n\}$
  FOR $j = 1, ..., n$ DO
  $\bar{c_{ij}} := c_{ij} - \alpha_i$

FOR $j = 1, ..., n$ DO
  $\beta_j := \min\{\bar{c_{ij}} \mid i = 1, ..., n\}$
  FOR $i = 1, ..., n$ DO
  $c^*_{ij} := c_{ij} - \beta_j$

matrices from lecture? pages 40-41

It will be now described how one can design steps S2 and S3 of the branch and bound algorithm. Let X_i denote a subset of round trips. $b(X_i)$ denotes a lower bound for the length of the round trips in X_i. $C(X_i)$ denotes the cost matrix for X_i. Moreover, let $(S_k, S_l)(X_i)$ denote a pair of cities S_k and S_l which is assigned to X_i. G contains the set of all round trips. $b(G) := \gamma(C)$, $C(G) := C^*$. The pair of cities $(S_k, S_l)(G)$ should be chosen so that $C(G)_{kl} = 0$. In each run through step S2, one chooses a subset X of roundtrips with a pair $(S_k, S_\lambda)(X)$ of cities and a corresponding matrix $C(X)$. The set X is divided into

$$X_1 := \{x \in X \mid x_{\kappa\lambda} = 1\}$$
$$X_2 := \{x \in X \mid x_{\kappa\lambda} = 0\}$$

The cost matrices for these subsets are given by

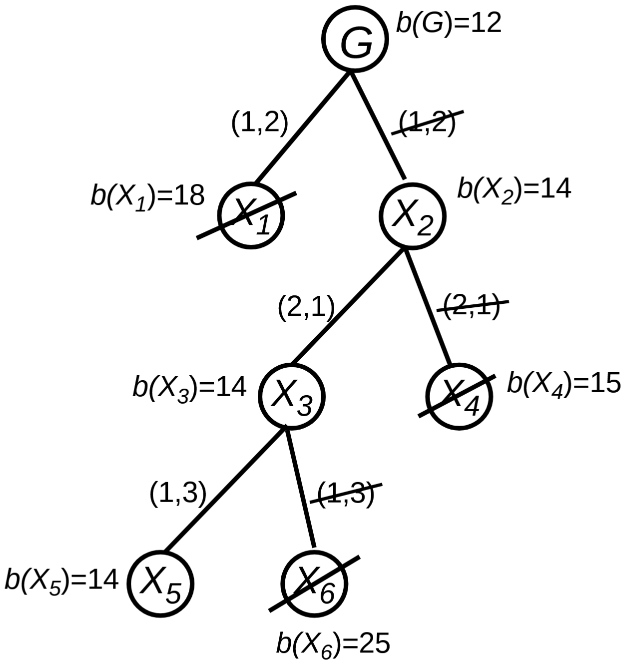
$$\tilde{C}(X_1)_{ij} := \begin{cases} C(X_1)_{ij}, & [i \neq \kappa \wedge j \neq \lambda] \text{ or } (i,j) = (\kappa, \lambda) \\ \infty, & \text{otherwise} \end{cases}$$

and $C(X_1) := \tilde{C}^*(X_1)$

$$\widehat{C}(X_2)_{ij} := \begin{cases} C(X)_{ij}, & (i,j) \neq (\kappa, \lambda) \\ \infty, & (i,j) = (\kappa, \lambda) \end{cases}$$

and $C(X_2) := \widehat{C}^*(X_2)$. Lower bounds for the sets X_1 and X_2 are given by

$$b(X_1) := b(X) + \gamma(\tilde{C}(X_1)), \quad b(X_2) := b(X) + \gamma(\widehat{C}(X_2))$$



X_5 enthält nur eine Rundreise, die auch optimal ist

Remark 2.29.

- One has to exclude all cases, where a subset of roundtrips is empty. Such a situation can occur if, due to ∞-costs, no roundtrip with finite costs exists in a subset of roundtrips (maybe there are only subtours).
- In order to exclude subtours, one can set the corresponding cost value in $\tilde{C}(X)$ or $\widehat{C}(X)$.
- The branching of a set X of roundtrips has been done so that (k, l) is chosen in a way that the connection $(S_k, S_l)(X_1)$ (i.e. $(S_k, S_l)(X_2)$) requires as least as possible costs $C(X_1)_{kl}$ ($C(X_2)_{kl}$ resp.)
- The way for reducing costs used here is quite simple and can be replaced by a more sophisticated technique ($\rightarrow$ exercise task).

2.4.5 The Method Branch & Cut

$$\min c^T x \quad \text{such that } x \in G := Q \cap \mathbb{Z}^n, \tag{ILP}$$

Algorithm 2: Branch & Cut

S1: $\mathcal{G} := \{G\}$, determine $y \in G$ and set $\widehat{b} := c^T y$ and set $Q(G) := Q$.

S2: (BRANCH) IF Step S2 is used for the first time, set $r := 1$. Otherwise, set $r \geq 2$. Choose $X \in \mathcal{G}$ and nonempty sets $X_1, \dots, X_r$ so that $X_i = Q(X_i) \cap \mathbb{Z}^n$, $\bigcup_{1 \leq i \leq r} X_i = X$ and $X_i \cap X_j = \emptyset$ for all i, j with $i \neq j$, where $Q(X_i)$ is a polyhedron defined by explicit linear inequalities.

S3: (BOUND) FOR $i = 1, ..., r$ DO

(a) COMPUTE a solution x^i of the continuous linear program:

$$\min_{x \in Q(X_i)} c^T x$$

Set $b(X_i) = c^T x^i$.
IF $x^i \in \mathbb{Z}^n$ THEN $X_i := \{x^i\}$ and GOTO (b).
IF $x^i \notin \mathbb{Z}^n$ THEN
(CUT) Generate an inequality $[p, p_0]$ that is valid for $\operatorname{conv}(Q(X_i) \cap \mathbb{Z}^n)$ such that $p^T x^i - p_0 > 0$.
$Q(X_i) := \{x \in Q(X_i) \mid p^T x - p_0 \leq 0\}$
IF "Test" is satisfied THEN GOTO (a)
ELSE GOTO (b)

(b) IF $i = r$ THEN GOTO S4
END FOR

S4: $\mathcal{G} := \mathcal{G} \setminus \{X\}$ and $\mathcal{G} := \mathcal{G} \cup \{X_1, ..., X_r\}$.
FORALL $Y \in \mathcal{G}$ DO
DETERMINE $z \in Y$. If $c^T z \leq \widehat{b}$ THEN $\widehat{b} := c^T z$ and $y := z$.
END FORALL.
FOR ALL $Y \in \mathcal{G}$ DO
IF $b(Y) \geq \widehat{b}$ THEN $\mathcal{G} := \mathcal{G} \setminus \{Y\}$.
END FORALL.
IF $\mathcal{G} \neq \emptyset$ THEN GOTO S2. ELSE $x^* := y$ and STOP.

Remark 2.30.

- If in Step S2 r is always set to 1, we would get a cutting plane method which work without branching. This method can be obtained as well if the "Test" in S3 (Cut) is always satisfied.
- It is always possible to cut of a noninteger solution x^i by a valid inequality for $\operatorname{conv}(Q(X_i \cap \mathbb{Z}^n)$. (To this end, one can solve the linear program in S3 (a) by the simplex method and use Gomory cuts.)
- By means of "Test" one can stop using poor cuts

3 Optimization in Graphs

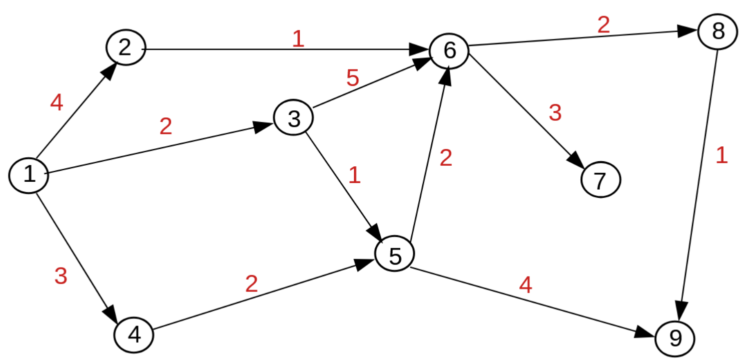
3.1 Basic Notations

Definition 3.1 (Graphs).
Let F be a nonempty set and $E \subseteq F \times F$. Then $G = (F, E)$ is called a directed graph (digraph) with vertex set F and edge set E.
If $f, g \in F$ and $(f, g) \in E$, then the edge (f, g) connects the vertices f and g. The edge (f, g) starts in f and ends in g.
If, for all $e \in E$, there exist weights $w_e \in \mathbb{R}$, then the triple $G = (F, E, (w_e)_{e \in E})$ is called weighted digraph.
For $U, V \subseteq F$ let $E(V, U) := E \cap (V \times U)$. Moreover:

$$E^+(V) := E(V, F \setminus V) \quad \text{(outer incident edges of } V\text{)}$$
$$E^-(V) := E(F \setminus V, V) \quad \text{(inner incident edges of } V\text{)}$$

$F^+(V) := \{g \in F \setminus V \mid \exists f \in V : (f, g) \in E\}$ is called set of outer adjacent vertices (set of successor vertices of V).
$F^-(V) := \{g \in F \setminus V \mid \exists f \in V : (g, f) \in E\}$ is called set of predecessors of V.
If $V = \{v\}$, we define $E^+(v) := E^+(\{v\})$, $E^-(v) := E^-(\{v\})$, $F^+(v) := F^+(\{v\})$ and $F^-(v) := F^-(\{v\})$.

Consider the following graph:



It holds:

$$
\begin{aligned}
F &= \{1, ..., 9\} \\
E &= \{(1,2), (1,3), (1,4), (2,6), (3,5), (3,6), (4,5), (5,6), (5,9), (6,7), (6,8), (8,9)\} \\
E^+(\{1,2\}) &= \{(1,3), (1,4), (2,6)\} \\
E^+(3) &= \{(3,5), (3,6)\} \\
E^-(6) &= \{(2,6), (3,6), (5,6)\} \\
F^+(\{3,5\}) &= \{6, 9\} \\
F^+(7) &= \emptyset \\
F^-(9) &= \{5, 8\}.
\end{aligned}
$$

The adjacence matrix $(\mathrm{Ad})_G \in \mathbb{R}^{|F| \times |F|}$ of a digraph $G = (F, E)$ is given by

$$(\mathrm{Ad}_G)_{i,j} = \begin{cases} 1, & \text{if edge starts in } i \text{ and ends in } j \\ 0, & \text{otherwise.} \end{cases}$$

The incidence matrix $(\mathrm{In})_G \in \mathbb{R}^{|F|\times|E|}$ of a digraph $G = (F, E)$ is given by

$$(\mathrm{In}_G)_{i,j} = \begin{cases} 1, & \text{if edge } j \text{ starts in vertex } i \\ -1, & \text{if edge } j \text{ ends in vertex } i \\ 0, & \text{otherwise} \end{cases}$$

We assume that $(i, j) \notin E$ for all $i \in F$ (the graph has no loops).

$$\mathrm{Ad}_G = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathrm{In}_G = \ldots$$

3.2 Shortest Paths in Graphs

Definition 3.2 (Path).
Let $G = (F, E)$ be a digraph, $n \in \mathbb{N}$ and $W = (e_1, \ldots, e_n) \in E^n = E \times \cdots \times E$.
For $e_i = (f_i, g_i)$ and $e_{i+1} = (f_{i+1}, g_{i+1})$ it holds $g_i = f_{i+1}$ for $i = 1, \ldots, n-1$.
We then say that the path W connects the vertex f_1 with the vertex g_n.
Let G be weighted by means of weights $(w_e)_{e\in E}$. Then the length of the path W is given by

$$\sum_{i=1}^{n} w_{e_i}.$$

A circle is a path $W = (e_1, \ldots, e_n)$ such that $e_1 = (f_1, g_1)$, $e_n = (f_n, g_n)$, and $f_1 = g_n$.

Algorithm 3: Dijkstra

S1 (Input): weighted digraph $G = (F, E, (w_e)_{e\in E})$ with $w_e \geq 0$, source vertex a, destination vertex b.
S2: FORALL $f \in F$ DO $d(f) := +\infty$, $p(f) := f$. END FORALL.
$A := \{a\}$, $B := F^+(a)$, $f := a$, $d(a) := 0$.
S3: WHILE $b \notin A$ AND $B \neq \emptyset$ DO

(a) (Update)
FOR ALL $g \in F^+(f)$ DO
IF $d(f) + w_{(f,g)} < d(g)$ THEN $d(g) := d(f) + w_{(f,g)}$, $p(g) = f$.

(b) (Determine the next edge on the shortest path)
CHOOSE $f \in B$ with $d(f) = \min\{d(g) \mid g \in B\}$

(c) $A := A \cup \{f\}$, $B := (B \cup F^+(f)) \setminus A$

S4: Output: Distance $d(b)$ between a and b.

table script page 51.

3.3 Maximal Flow

$G = (F, E, (w_e)_{e\in E})$ is a given digraph.

3.3.1 Basic Notation and Model

$Q, S \subset F$ with $Q \cap S = \emptyset$ and $E^-(Q) = \emptyset = E^+(S)$.
$E^+(V) = E(V, F \setminus V)$, $E^+(S) = E(S, F \setminus S)$, $E^-(Q) = E(F \setminus Q, Q)$.
$F \setminus (Q \cup S) =$ intermediate vertices.
Instead of Q, we introduce unique source vertex (artificial source) $\widehat{q}$.
Instead of S, we introduce a unique sink vertex (artificial sink) $\widehat{s}$.
For each $e \in E$, let $w_e \geq 0$ be the maximal capacity. We assume that the minimal capacity of each edge $e \in E$ is 0.
Modeling: Let $x_e \in [0, \infty)$ be variables for $e \in E$ representing the flow on edge e.

$$\max \sum_{e\in E(Q,F\setminus Q)} x_e \quad \text{such that} \tag{44}$$

$$\sum_{e\in E(f,F)} x_e - \sum_{e\in E(F,f)} x_e = 0 \quad f \in F \setminus (Q \cup S) \tag{45}$$

$$0 \leq x_e \leq w_e \tag{46}$$

(45) is called Kirchoff's vertex equations. We assume WLOG that $Q = \{\widehat{q}\}$ and $S = \{\widehat{s}\}$.

image

Instead of (44)-(46), we now use the model

$$\max c^T x \quad \text{such that} \quad Ax = 0,\ 0 \leq x \leq w. \tag{47}$$

with

$$c = (c_e)_{e\in E} \quad \text{with} \quad c_e = \begin{cases} 1, & e = (\widehat{s}, \widehat{q}) \\ 0, & \text{otherwise.} \end{cases}$$

and

$$A = \mathrm{In}_G \in \mathbb{R}^{|F|\times|E|}.$$

$$Ax = 0 \iff \text{model for the problem of the maximal flow in } G.$$

If we write the constraints in (47) as

$$\begin{aligned} (Ax \leq 0) \\ (Ax \geq 0) \\ x \geq 0 \\ x \leq w \end{aligned}$$

By the Weierstrass theorem, this problem is solvable (since $x = 0$ is feasible, the feasible region is compact, and the objective function is continuous).
A vector x satisfying all constraints of problem (47) is called a *flow*. For a flow x, we denote $x_{(s,q)} = c^T x$ as the *flow strength* or *value* of the flow.
Writing the constraints of (47) in the form $\tilde{A}x \leq \tilde{b}, x \geq 0$, we can apply Theorem 2.24 and Lemma 2.23 to show total unimodularity of $\tilde{A}$. Thus, the simplex method could compute an optimal vertex of problem (47) that is automatically integer (provided $\tilde{b}$ is integer). This approach is particularly useful when imposing integrality constraints ($x \in \mathbb{Z}^{|E|}$). However, we will not exploit total unimodularity in the following.

3.3.2 Minimal Cut - Maximal Flow

The dual problem to (47) reads as follows

$$\min -w^T v \qquad \text{subject to} \qquad A^T u + v \leq -c, v \leq 0 \tag{48}$$

Since the primal linear optimization problem (47) is solvable, the corresponding dual problem (48) must also have a solution. We identify the vector $u \in \mathbb{R}^{|F|}$ with the vector $(u_f)_{f\in F}$ and we identify $c, v, w \in \mathbb{R}^{|E|}$ with the vectors $(c_e)_{e\in E}, (v_e)_{e\in E}, (w_e)_{e\in E}$. Taking into account the structure of the incidence matrix A and the definition of the vector c, then (48) can also be written as

$$\min -\sum_{e\in E} w_e v_e \tag{49}$$

$$\text{subject to } u_s - u_q + v_{(s,q)} \leq -1 \tag{50}$$

$$u_f - u_g + v_e \leq 0 \qquad \text{for } e = (f,g) \in E \setminus \{(s,q)\} \tag{51}$$

$$v_e \leq 0 \qquad \text{for } e \in E \tag{52}$$

Theorem 3.3.
Let $V \subset F$ be a node set with $s \in V$ and $q \notin V$. Then (u, v) with

$$u_f := \begin{cases} 0, & \text{if } f \in V \\ 1, & \text{if } f \in F \setminus V \end{cases}$$

$$v_e := \begin{cases} -1, & \text{if } e = (f,g) \in E(F\setminus V, V) \\ 0, & \text{otherwise} \end{cases}$$

is feasible for (48).

Proof. Check conditions (50)-(51). □

Definition 3.4 (Cuts).
A node set $V \subset F$ with $s \in V$ and $q \notin V$ is called a <u>cut</u> of the weighted digraph G. The value

$$\sum_{e\in E(F\setminus V,V)} w_e$$

is called <u>the capacity of the cut</u>.

We recognize that for every cut V, Theorem 3.3 provides a (u, v) that is feasible for problem (48). The objective function value can be easily determined taking into account (49) and Theorem 3.3:

$$-w^T v = -\sum_{e\in E} w_e v_e = \sum_{e\in E(F\setminus V,V)} w_e$$

This is precisely the capacity of the cut V. One therefore suspects that the dual problem (48) seeks to find a cut with minimal capacity.

Theorem 3.5 (Max Flow-Min Cut; Ford/Fulkerson).
In the flow problem (47):

1. Value of maximal flow = optimal value of dual problem (48) = minimal cut capacity.
2. If V is a minimal cut and $(x_e)_{e\in E}$ is maximal flow, then $x_e = w_e$ for all $e \in E(F \setminus V, V)$.

3.3.3 The Algorithm of Ford-Fulkerson

$(w_e)_{e\in E} \geq 0$, one source q, one sink s, $(s,q) \in E$. WLOG assume that, for each $(f,g) \in E \setminus \{s,q\}$, (g,f) belongs to E as well with $w_{(g,f)} := 0$. Moreover, we exclude loops (f,f).

Definition 3.6.
For $x = (x_e)_{e\in E}$ with $0 \leq x_e \leq w_e$ for all $e \in E$. The residuum of an edge $e = (f,g) \in E$ is given by

$$r(e) := w_e - x_e + x_{(g,f)} \quad \text{for } e \neq (s,q) \quad r((s,q)) := w_{(s,q)} - x_{(s,q)}.$$

By

$$R(x) := \{e \in E \mid r(e) > 0\}$$

the residual graph is defined by

$$G(x) := (F, R(x))$$

Algorithm 4: Ford-Fulkerson-Algorithm, Augmenting Path Algorithm

S1 (Input): Graph $G = (F, E, (w_e)_{e\in E})$ with $w_e \geq 0$ for all $e \in E$, source q, sink s.
S2: $x = (x_e)_{e\in E} := 0$
S3: Determine the residual graph $G(x)$ and the distance $d(x)$ between q and s (w.r.t. the number of edges)
S4: WHILE $d(x) < +\infty$ DO

(a) Determine a path $W = W(x) = (e_1, ..., e_n)$ which connects q and s within $G(x)$.

(b) $\delta := \min\limits_{1\leq i\leq n} r(e_i)$

(c) (Augmenting step)
FOR $i = 1, ..., n$ DO

(i) $e := e_i =: (f,g)$

(ii) IF $\delta \leq w_e - x_e$ THEN $x_e := x_e + \delta$
ELSE $y := x_e$, $x_e := w_e$, $x_{(g,f)} := x_{(g,f)} - \delta + w_e - y$

(d) Compute $G(x)$ and the distance $d(x)$ between q and s in $G(x)$. END WHILE

S5: $x_{(s,q)} := \sum\limits_{e\in E(q,F)} x_e$

Theorem 3.7.
If Algorithm 4 terminates (condition in the WHILE-loop in S4 is not satisfied), then the computed flow x is maximal.

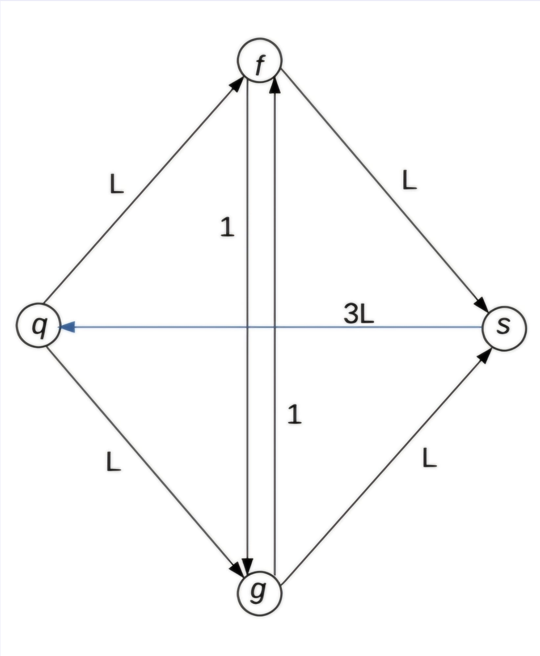
Proof. Exercise. □

Theorem 3.8.
Let all weights of the graph G be integers. Furthermore, let $B := \sum_{e \in E(q,F)} w_e$. Then the Ford-Fulkerson algorithm terminates in S6, where the WHILE-loop in S4 was executed at most B times.

Proof. See page 56. □

If the edge weights are allowed to be irrational, there exist examples where Algorithm 4 (with exact computation) does not terminate.

Example 3.9.
Consider



$$\begin{aligned}
&L \geq 1 \\
&F = \{q, s, f, g\} \\
&E = \{(q, f), (q, g), (f, g), (g, f), (f, s), (g, s), (s, q)\} \\
&w_{fg} = w_{gf} = 1, \quad w_{qf} = w_{qg} = w_{fs} = w_{gs} = L, \quad w_{sq} = 3L
\end{aligned}$$

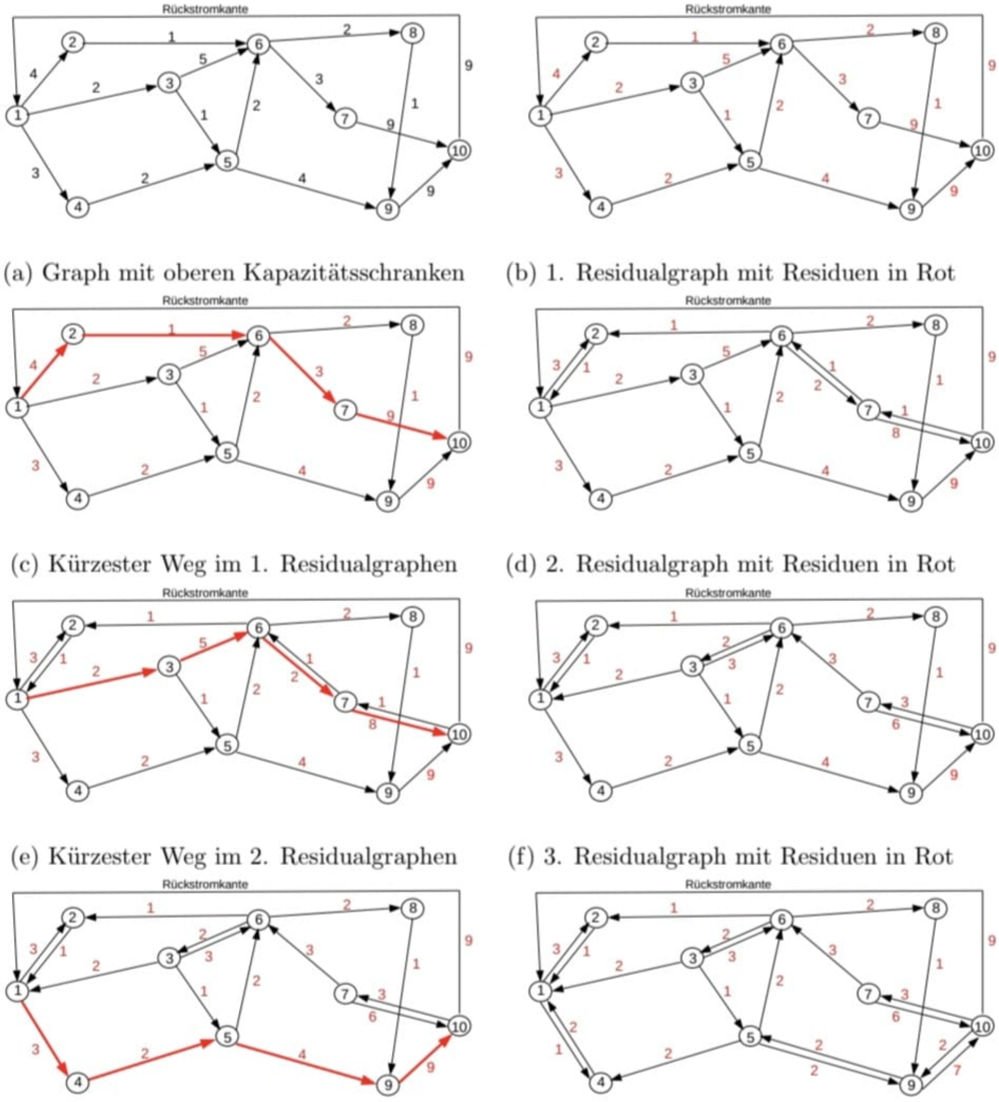
For simplicity, edges with capacity 0 have been omitted. When executing Algorithm 4, some path W must be constructed that connects q with s in the residual graph $G(x)$.
1) If this path alternates between $((q, f), (f, g), (g, s))$ and $((q, g), (g, f), (f, s))$, then in

each iteration of the WHILE-loop the value of the flow is increased by $\delta = \min\{L, 1, L\}$ $= 1$. Thus Algorithm 4 requires $2L$ iterations of the WHILE-loop until termination.
2) If, on the other hand, in Algorithm 4 the paths $((q, f), (f, s))$ and $((q, g), (g, s))$ are generated consecutively, then the maximal flow of $2L$ is reached after two executions of the WHILE-loop, so the algorithm terminates after only two executions of the WHILE-loop.

The improvement according to Dinic or Edmonds and Karp
In Algorithm 4, instruction S4 (a) is replaced by
S4 $(a)'$ Compute a shortest path $W = W(x) = (e_1, ..., e_n)$ (with respect to the number of edges) that connects node q with s in $G(x)$.

Theorem 3.10.
Algorithm 4 with the improvement according to Dinic or Edmonds and Karp executes the WHILE-loop in S4 at most $|F|\frac{|E|}{2}$ times.

Proof. Without proof. □



(a) Graph mit oberen Kapazitätsschranken (b) 1. Residualgraph mit Residuen in Rot

(c) Kürzester Weg im 1. Residualgraphen (d) 2. Residualgraph mit Residuen in Rot

(e) Kürzester Weg im 2. Residualgraphen (f) 3. Residualgraph mit Residuen in Rot

(g) Kürzester Weg im 3. Residualgraphen (h) 4. Residualgraph mit Residuen in Rot

Abbildung 9: Beispiel zur Bestimmung eines maximalen Flusses

Value of the maximal flow = 5

4 Greedy-Algorithms and Matroids

Our optimization problem in this chapter:

$$\min_{M \in \mathcal{M}} c(M) \tag{58}$$

with $\mathcal{M} \subseteq \mathcal{P}(N)$, where $\mathcal{P}(N)$ denotes the power set of $N := \{1, ..., n\}$, and the objective function $c : \mathcal{P}(N) \to \mathbb{R}$, where $c(\{\ell\})$ are given for all $\ell \in \mathbb{N}$, $c(\emptyset) := 0$, and

$$c(M) := \sum_{\ell \in M} c(\{\ell\}) \qquad \forall M \in \mathcal{P}(N).$$

This objective function is called modular cost function.
Knapsack problem: $\max c^T x$ such that $a^T x - a_0 \leq 0$, $x \in \{0,1\}^n$ with given $a, c \in \mathbb{Z}_+^n$, $a_0 \in \mathbb{Z}_+$.

This knapsack problem can be written in the form of (58) by setting $\mathcal{M} := \left\{ M \subset N \mid \sum_{\ell \in M} a_\ell \leq a_0 \right\}$ and $c(\{\ell\}) = -c_\ell$ for all $\ell \in N$.
Determining a minimal spanning tree.
Let $G := (F, N, (w_i)_{i \in N})$ be an undirected connected graph with nonnegative weights w_i for all $i \in N$. The graph is called connected, if, for any two vertices $u, v \in F$, there exists a path of edges in N so that this path connects u and v. A spanning tree of G is a connected circle free graph (F, N') with $N' \subseteq N$. The value of a spanning tree is given by $\sum_{i \in N'} w_i$.
Thus, a minimal spanning tree of G is a graph (F, N^*) with $N^* \subseteq N$ such that (F, N^*) is a spanning tree with minimal value, i.e. $\sum_{\ell \in N^*} w_\ell \leq \sum_{\ell \in N'} w_\ell$ for all spanning trees (F, N') of G. To reformulate this problem in the form of (58), we set

$$\mathcal{M} := \{M \subseteq N \mid (F, M) \text{ is circle-free}\}$$

and

$$c(\{\ell\}) := w_\ell - (\max\{w_i \mid i \in N\} + 1) \quad \text{ for each } \ell \in N.$$

With this, every solution of (58) yields a minimal spanning tree of G. Vice versa, the edge set of a minimal spanning tree of G solves (58) ($\to$ Exercises).

Algorithm 5: Greedy Algorithm

S1: The ordering

$$c(\{1\}) \leq c(\{2\}) \leq \cdots \leq c(\{n\}) \leq 0 \tag{59}$$

is satisfied. Set $M := \emptyset$.
S2: FOR $i = 1, \ldots, n$ DO
IF $M \cup \{i\} \in \mathcal{M}$ THEN $M := M \cup \{i\}$
S3: $M^* := M$ and stop.

In general, one cannot expect that M^* from S3 is an optimal solution of (58). To prove that M^* is an optimal solution for "all" choices of c, one can state a condition on $\mathcal{M}$.

Definition 4.1 (Independent System and Matroid).

Let $N := \{1, \ldots, n\}$ and $\mathcal{M} \subseteq \mathcal{P}(N)$. The pair $(N, \mathcal{M})$ is an *independent system* if:

$$\emptyset \in \mathcal{M} \tag{M1}$$
$$\forall L, M \subseteq N : [L \subseteq M \in \mathcal{M} \implies L \in \mathcal{M}] \tag{M2}$$

Elements of $\mathcal{M}$ are *independent*, elements of $\mathcal{P}(N) \setminus \mathcal{M}$ are *dependent*.
$M \in \mathcal{M}$ is *maximally independent* (a *basis*) if $M \cup \{\ell\} \notin \mathcal{M}$ for all $\ell \in N \setminus M$.
$M \subseteq N$ is a *circuit* if $M \notin \mathcal{M}$ and $M \setminus \{\ell\} \in \mathcal{M}$ for all $\ell \in M$.
An independent system $(N, \mathcal{M})$ is a *matroid* if:

$$\forall L, M \in \mathcal{M} : [|M| = |L| + 1 \implies \exists a \in M \setminus L : L \cup \{a\} \in \mathcal{M}] \tag{M3}$$

Definition 4.2.
Let $(N, \mathcal{M})$ be an Independent System. For an arbitrary $T \subseteq N$

$$r(T) := \max\{|S| : S \subseteq T, S \in \mathcal{M}\}$$

is called rank of T and

$$\rho(T) := \min\{|S| \mid S \subseteq T, S \in \mathcal{M}, S \cup \{t\} \notin \mathcal{M} \quad \forall t \in T \setminus S\}$$

is called lower rank of T. The rank quotient of a non-trivial Independent System $(N, \mathcal{M})$ is defined as

$$q(N, \mathcal{M}) := \min \left\{ \frac{\rho(T)}{r(T)} : r(T) \neq 0,\ T \subseteq N \right\}.$$

Obviously, for any Independent System $(N, \mathcal{M})$ it holds that $q(N, \mathcal{M}) \leq 1$.

Lemma 4.3.
Let $(N, \mathcal{M})$ be a non-trivial Independent System. Then TFAE:

$$\forall L, M \in \mathcal{M} : [|M| = |L| + 1 \implies \exists a \in M \setminus L : L \cup \{a\} \in \mathcal{M}] \tag{M3}$$
$$\forall L, M \in \mathcal{M} : [|M| > |L| \implies \exists a \in M \setminus L : L \cup \{a\} \in \mathcal{M}\} \tag{M3'}$$
$$q(N, \mathcal{M}) = 1. \tag{M3''}$$

Proof. (M3”) $\implies$ (M3): Assume that (M3) is wrong. Then there exist $L, M \in \mathcal{M}$ with $|M| = |L| + 1$ so that $L \cup \{a\} \notin \mathcal{M}$ for all $a \in M \setminus L$. Because of $(L \cup M) \setminus L = M \setminus L$, it follows

$$\rho(L \cup M) = \min\{|S| : S \subseteq L \cup M,\ S \in \mathcal{M},\ S \cup \{a\} \notin \mathcal{M} \quad \forall a \in (L \cup M) \setminus S\} \leq |L|$$

by taking $S = L$. On the other hand, it holds that

$$r(L \cup M) \geq \max\{|L|, |M|\} = |L| + 1 > |L|.$$

This contradicts $q(N, \mathcal{M}) = 1$.
(M3) $\implies$ (M3’): Exercise.
(M3’) $\implies$ (M3”): Let us assume that (M3”) is not satisfied. Then there exist $T \subseteq N$, so that $\rho(T) < r(T)$. Therefore, there are sets $L \subseteq T$ and $M \subseteq T$ so that $L, M \in \mathcal{M}$ and

$$|L| = \rho(T) < r(T) = |M| \quad \text{and} \quad L \cup \{b\} \notin \mathcal{M}\ \forall b \in T \setminus L.$$

This is a contradiction to (M3’) and, hence, the implication (M3’) $\implies$ (M3”) must hold. $\square$

Theorem 4.4.
For an arbitrary non-trivial independent system $(N, \mathcal{M})$ and an arbitrary modular continuous function $c : \mathcal{P}(N) \to (-\infty, 0]$, let $OPT(N, \mathcal{M}, c)$ denote the optimal objective function value of problem (58). Moreover, let $G(N, \mathcal{M}, c)$ denote the function value $c(M^*)$ of the set M^* which is computed by the Greedy Algorithm (Algorithm 5). If $OPT(N, \mathcal{M}, c) < 0$, then it holds that

$$q(N, \mathcal{M}) \leq \frac{G(N, \mathcal{M}, c)}{OPT(N, \mathcal{M}, c)} \leq 1.$$

In particular, there is at least one cost function c for which the quotient is equal to the lower bound $q(N, \mathcal{M})$.

Proof. WLOG, we assume $c(\{1\}) \leq \cdots \leq c(\{n\})$. Because of $c : \mathcal{M} \to (-\infty, 0]$, we have $c(\{i\}) \leq 0$ for all $i \in N$ and thus (59) is satisfied. Now, the following sets are defined:

$$\begin{aligned}
&G_n := M^* \\
&O_n \in \arg\min\{c(M) : M \in \mathcal{M}\} \quad \text{(a solution of (58))} \\
&N_0 := \emptyset \\
&O_0 := \emptyset \\
&N_J := \{1, ..., j\} \\
&G_j := G_n \cap N_j \\
&O_j := O_n \cap N_j,
\end{aligned}$$

where $j = 0, ..., n$. For any of these j it follows from (M2) that $O_j \in \mathcal{M}$, $O_j \subseteq N_j$ and $|O_j| \leq r(N_j)$. According to the Greedy Algorithm it holds for all $j \in \{0, 1, ..., n\}$ that $G_j \in \mathcal{M}$ and $G_j \cup \{t\} \notin \mathcal{M}$ for all $t \in N_j \setminus G_j$. Hence, $|G_j| \geq \rho(N_j)$. With $d_j := c(\{j\}) - c(\{j+1\})$ for $j = 1, ..., n-1$ and $d_n := c(\{n\})$, we get

$$\begin{aligned}
G(N, \mathcal{M}, c) = c(G_n) &= \sum_{j=1}^{n} (|G_j| - |G_{j-1}|) c(\{j\}) \\
&= \sum_{j=1}^{n} |G_j| d_j \\
&\geq \sum_{j=1}^{n} \rho(N_j) d_j && \text{because } d_j \leq 0 \text{ for } j = 1, \ldots, n \\
&\geq q(N, \mathcal{M}) \sum_{j=1}^{n} r(N_j) d_j && \text{as } \rho(N_j) \geq q(N, \mathcal{M}) r(N_j) \text{ for } j = 1, \ldots, n \\
&\geq q(N, \mathcal{M}) \sum_{j=1}^{n} |O_j| d_j \\
&= q(N, \mathcal{M}) \sum_{j=1}^{n} (|O_j| - |O_{j-1}|) c(\{j\}) \\
&= q(N, \mathcal{M}) c(O_n) \\
&= q(N, \mathcal{M}) OPT(N, \mathcal{M}, c).
\end{aligned}$$

See the exercises for a modular cost function attaining $q(N, \mathcal{M})$. □

Theorem 4.5.
A non-trivial independence system is a matroid $\iff$ the Greedy Algorithm provides a solution of problem (58) for each modular cost function $c : \mathcal{P}(N) \to (-\infty, 0]$ with $OPT(N, \mathcal{M}, c) < 0$.

Proof. According to Theorem 4.4 it holds that $q(N, \mathcal{M}) < 1$ if and only if there exists a modular cost function for which the Greedy Algorithm does not provide a solution of (58). By means of Lemma 4.3, we know that $q(N, \mathcal{M}) < 1$ if and only if $(N, \mathcal{M})$ is not a matroid. □

Theorem 4.6 (without proof).
Let $(N, \mathcal{M})$ be an independence system. If the set $M \cup \{a\}$ for each $M \in \mathcal{M}$ and each $a \in N$ contains at most p circuits, then

$$q(N, \mathcal{M}) \geq \frac{1}{p}.$$

$G = (F, N, (w_i)_{i \in N})$ undirected and connected graph. It will be shown in the Exercises that the following algorithm can be regarded as a Greedy Algorithm in the sense of Algorithm 5.

Algorithm 6: Kruskal's Algorithm

S1: Let the ordering $w_1 \leq w_2 \leq \cdots \leq w_n$ be given. $H := \emptyset$.
S2: FOR $i = 1, ..., n$ DO IF $(F, H \cup \{i\})$ does not contain a circle then set $H := H \cup \{i\}$.
S3: $G^* = (F, H)$ is a minimal spanning tree of G, STOP.

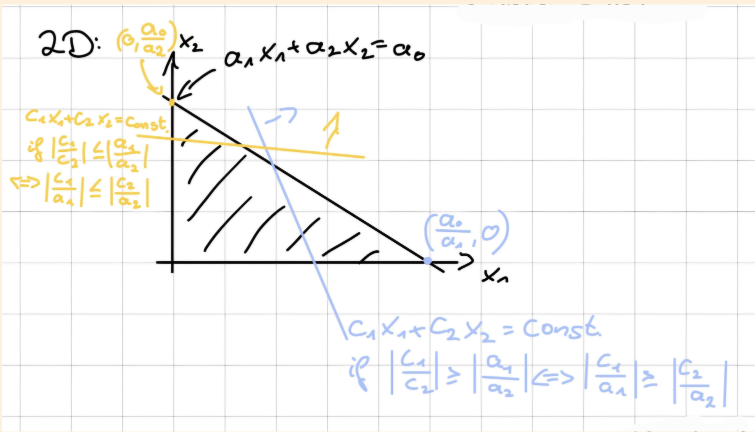
A Tutorials

A.1 Tutorial 1

Exercise 1 (Continuous and Discrete Knapsack Problems).

(a) Knapsack problem:

$$c_1x_1 + \cdots + c_nx_n \to \max \quad \text{such that} \begin{cases} a_1x_1 + \cdots + a_nx_n \leq a_0 \\ x \in \mathbb{R}^n_+ \end{cases}$$



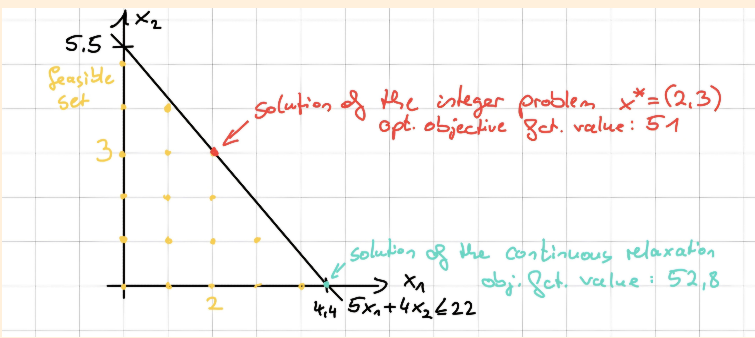
In general: Let i_0 be an index with $\left|\frac{c_{i_0}}{a_{i_0}}\right| \geq \left|\frac{c_i}{a_i}\right|$ for all $i = 1, ..., n$ and define x^* by $x_i^* := \begin{cases} \frac{a_0}{a_{i_0}}, & i = i_0 \\ 0, & i \neq i_0. \end{cases} \implies x^*$ is a solution because

$$c^Tx^* = \sum_{i=1}^n c_ix_i^* = c_{i_0}\frac{a_0}{a_{i_0}}$$

For any feasible point:

$$c^Tx = \sum_{i=1}^n c_ix_i = \sum_{i=1}^n \underbrace{\frac{c_i}{a_i}}_{\leq \frac{c_{i_0}}{a_{i_0}}} \leq \frac{c_{i_0}}{a_{i_0}} \cdot \underbrace{\sum_{i=1}^n a_ix_i}_{\leq a_0} \leq c^Tx^*.$$

(b) We can draw the problem to solve it graphically



Exercise 2 (Solvability of Discrete Optimization Problems and their Continuous Relaxation).

$$c_1x_1 + c_2x_2 \to \min \quad \text{such that} \begin{cases} a_1x_1 + a_2x_2 + a_0 = 0 \\ x_1, x_2 \in \mathbb{Z}_+. \end{cases}$$

(a) Example:

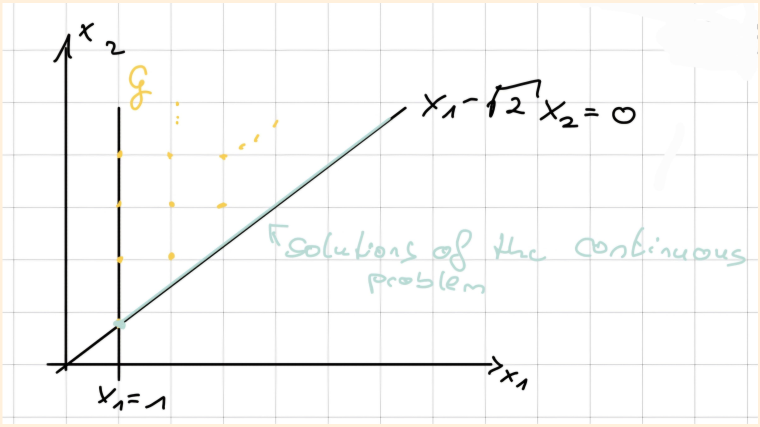
$$-x_1 \to \min \quad \text{such that} \begin{cases} x_1 = \sqrt{2}x_2, \\ x_1, x_2 \in \mathbb{Z}_+ \end{cases}$$

(b) Example:

$$x_1 \to \min \quad \text{such that} \begin{cases} x_2 = \frac{1}{\sqrt{x_2}}x_1 + \frac{1}{2}(\iff x_1 - \sqrt{2}x_2 + \frac{\sqrt{2}}{2} = 0) \\ x_1, x_2 \in \mathbb{Z}_+ \end{cases}$$

Exercise 3 (Solvability of Integer Optimization Problems with Irrational Input Data).

$$f(x) := -x_1 + \sqrt{2}x_2 \to \min \quad \text{such that} \begin{cases} x_1 - \sqrt{2}x_2 \leq 0 \\ -x_1 + 1 \leq 0 \\ x_1, x_2 \in \mathbb{Z}_+ \end{cases}$$



To show: $\inf_{x\in G} f(x) = 0$, but there is no $x \in G$ such that $f(x) = 0$ ($\implies$ integer problem does not have an optimal solution). Use hint for $r = \sqrt{2}$.

$$\implies \forall \varepsilon \in (0,1)\ \exists p, q \in \mathbb{Z}_+ \text{with } p \geq 1 : \underbrace{0 \leq \sqrt{2}q - p < \varepsilon}_{\iff 0 \leq f(p,q) < \varepsilon}$$

$$\iff \forall \varepsilon \in (0,1) \text{ there is } x \in G \text{ with } 0 \leq f(x) < \varepsilon$$

$$\iff \inf_{x\in G} f(x) = 0$$

Exercise 4. From lecture the UFL ist already known (see Section 1):

$$\sum_{s\in S} c_s x_s + \sum_{k\in K}\sum_{s\in S} d_{ks} y_{ks} \quad \text{such that} \quad \begin{cases} \cancel{\sum_{s\in S} y_{ks} = 1} & \forall k \in K \\ y_{ks} \leq \cancel{x_s}\ u_s x_s & \forall k \in K, s \in S \\ y_{ks} \geq 0 & \forall k \in K, s \in S \\ x_s \in \{0,1\} & \forall s \in S \\ \sum_{k\in K} y_{ks} & \\ \sum_{k\in K} y_{ks} \leq u_s & \forall s \in S \\ \sum_{s\in S} y_{ks} \geq b_k & \forall k \in K \end{cases}$$

- $K = \{1,\ldots,m\}$ set of customers
- $S = \{1,...,n\}$ set of possible facility locations
- variables $x_s = \begin{cases} 1, & \text{if facility is opened at location } s \\ 0, & \text{otherwise} \end{cases}$

  $y_{ks} \geq 0$ absolute amount of demand of customer k served from location s.

A.2 Tutorial 2

Polyhedron in $\mathbb{R}^n$:

$$P(C,d) = \{x \in \mathbb{R}^n \mid Cx \leq d\}$$

Exercise 1 (Basic Concepts of Polyhedral Theory).
Given $x^1 = (2,0,0)$, $x^2 = (0,5,0)$, $x^3 = (0,0,3)$. $X := \{x^1, x^2, x^3\}$.

(a)

$$\begin{aligned}\mathrm{aff}(X) &= X + \mathrm{lin}(X - X)\\ &= \{x^1, x^2, x^3\} + \mathrm{lin}\{x^2 - x^1, x^3 - x^1, x^3 - x^2\}\\ &= \{x^1 + \tau(x^2 - x^1) + s(x^3 - x^1) + \cancel{t(x^3 - x^2)},\ r,\ s,\ t \in \mathbb{R}\}\\ &\to \text{whole plane including the points } x^1, x^2, x^3\end{aligned}$$

Equation of the plane: $15_1 + 6x_2 + 10x_3 = 30$

(b) $\mathrm{conv}(X) \to$ triangle with vertices x^1, x^2, x^3.
From now, we consider the polyhedron $P := \mathrm{conv}(X)$.

(c)

$$\begin{aligned}x_1, x_2, x_3 &\geq 0\\ 15x_1 + 6x_2 + 10x_3 &\leq 30\\ 15x_1 + 6x_2 + 10x_3 &\geq 30\end{aligned}$$

(d)

$$\mathrm{rint}(P) = \{x \in P \mid \exists \varepsilon > 0 : (x + \varepsilon\mathcal{B}) \cap \mathrm{aff}(P) \subseteq P\}$$

In general: all points satisfy the inequalities $c_i^T x \leq d_i$ with "<", except the inequalities belonging to $I_=$.
Here: all points of the triangle except the points on the edges.

$$\mathrm{rbd}(P) = \mathrm{cl}(P) \setminus \mathrm{rint}(P) = P \setminus \mathrm{rint}(P) \to \text{ edges of the triangle}$$

(e) For the dimension of P we have

$$\begin{aligned}\dim(P) &= \dim(\mathrm{lin}(P - P))\\ &= \dim(\mathrm{lin}\{x^2 - x^1, x^3 - x^1\})\\ &= 2.\end{aligned}$$

(f) Reminder:

- $p^T x \leq p_0$ is valid inequality of P $\iff \forall x \in P : p^T x \leq p_0$
- face represented by valid inequality:

$$F = \{x \in P \mid p^T x = p_0\}$$

(a) $-x_1 \leq 0$ is valid inequality of P.
Corresponding face: $F_1 = \{x \in P \mid x_1 = 0\} \to$ connecting line between x^2 and x^3. $\dim(F_1) = 1$

(b) $x_2 \leq 5$ is valid inequality.
Corresponding face: $F_2 = \{x \in P \mid x_2 = 5\} = \{x^2\}$, $\dim(F_2) = 0$

(c) $x_3 \leq 4$ valid inequality.
$F_3 = \{x \in P \mid x_3 = 4\} = \emptyset$. $\dim(F_3) = -1$

(d) $2x_1 + 3x_2 \leq 6$ not valid inequality for P, for example x^2 does not satisfy it.

(e) $3x_1 + 2x_3 \leq 6$ valid inequality.
$F_5 = \{x \in P \mid 3x_1 + 2x_3 = 6\} \to$ connecting line between x^1 and x^3.

Exercise 2 (Affine Independence of Vectors).
$x^1, ..., x^k$ are called affinely independent if

$$\sum_{i=1}^{k} \lambda_i x^i = 0 \text{ and } \sum_{i=1}^{k} \lambda_i = 0$$

only possible for $\lambda_1 = \lambda_2 = ... = \lambda_k = 0$
<u>**(a) $\implies$ (b):**</u> To show: $x^1, ..., x^k$ affinely independent $\implies$ $x^2 - x^1, ..., x^k - x^1$ linearly independent.

$$\begin{aligned}
\sum_{i=2}^{k} \alpha_i (x^i - x^1) = 0 \iff & \underbrace{-\left(\sum_{i=2}^{k} \alpha_i\right)}_{=:\lambda_1} x^1 + \sum_{i=2}^{k} \underbrace{\alpha_i}_{=:\lambda_i} x^i = 0 \\
\implies & \sum_{i=1}^{k} \lambda_i x^i = 0 \text{ and } \sum_{i=1}^{k} \lambda_i = 0 \\
\implies & \lambda_1 = ... = \lambda_k = 0 \\
\implies & \alpha_2 = ... = \alpha_k = 0, \text{ i.e. } x^2 - x^1, ..., x^k - x^1 \text{ linearly independent.}
\end{aligned}$$

Proofs for (b) $\implies$ (a) $\iff$ (c) similar.

Exercise 3 (Affinely Independent Solutions of Linear Systems of Equations).
$Ax = b$. To show: $Ax = b$ has $n + 1 - \operatorname{rank}(A)$ affinely independent solutions, but not more.

Proof. Set $k := n - \operatorname{rank}(A)$. From La, it is known that $Ax = 0$ has k linearly independent solutions $x^1, ..., x^k$. Let $\widehat{x}$ be a solution of $Ax = b$.
$\implies$ $\widehat{x}, \widehat{x} + x^1, ..., \widehat{x} + x^k$ are $k + 1$ solutions of the system $Ax = b$, and they are affinely independent by Exercise 2.
Assume that $Ax = b$ has even $k + 2$ affinely independent solutions $y^1, ..., y^{k+2}$ $\implies$ $y^{k+2} - y^1, y^{k+1} - y^1, \ldots, y^2 - y^1$ are linearly independent <u>and</u> solutions of $Ax = 0$. This is a contradiction, as $Ax = 0$ has at most k linear independent solutions. $\square$

A.3 Tutorial 3

Exercise 1.

(a)

(b) $\inf\{c^T x \mid x \in Q_I\}$ finite $\implies$ (CILP) has (at least) a solution. Proof will be a done using a well-known result from linear optimization. Namely Theorem 2.6 in the german lecture notes on OPAL:

- feasible set Q_I nonempty, since G is nonempty,
- objective function is bounded from below on Q_I due to (*)

$\implies$ there is at least one vertex x^* of Q_I which solves (CILP), cf. Lemma 2.11 in the lecture notes. Let x^* be an optimal vertex of Q_I.

- $x^* \in \mathbb{Z}^n$ because all vertices of Q_I are integer $\implies$ x^* is feasible for (ILP).
- Assume there is $\widehat{x} \in G$ with $c^T\widehat{x} < c^T x^*$. But $G \subseteq Q_I \implies \widehat{x} \in G_I$, i.e. feasible for (CILP)
  $\implies$ contradiction, because x^* is a solution os (CILP).
  $\implies$ x^* is also a solution of (ILP).

(c) $\inf\{c^T x \mid x \in G\}$ finite. Using (a) in theorem 2.27, $\inf\{c^T x \mid x \in Q_I\}$ is finite, too. $\overset{(b)}{\implies}$ (CILP) is solvable and therefore (ILP) too. Let $x^* \in G$ be a solution of (ILP).

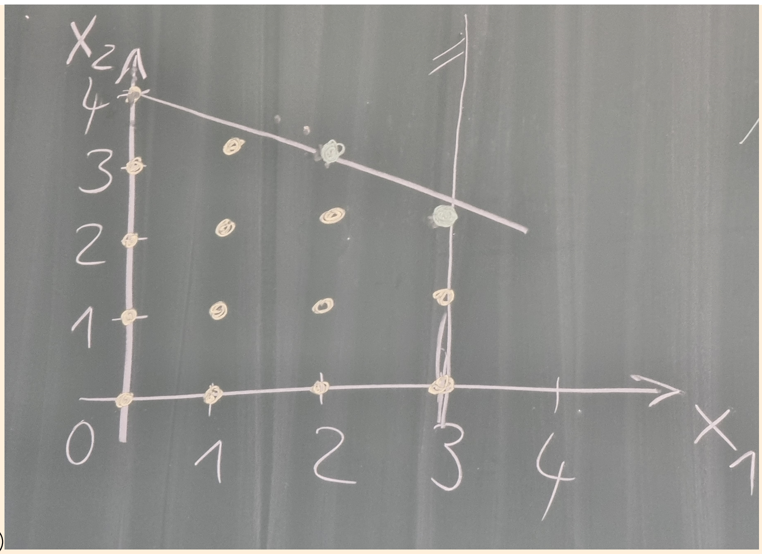
- $x^* \in Q_I$ because $G \subseteq Q_I$
- Assume there is some $\widehat{x} \in Q_I$ with $c^T\widehat{x} < c^T x^*$. WLOG, let $\widehat{x}$ be a vertex of $Q_I \implies \widehat{x} \in G$.

Exercise 2. Consider

$$\min_{x\in G} -x_1 - x_2 \quad \text{with } G := \{x \mid x_1 + 2x_2 \le 8, x_1 \le 3, x_1, x_2 \ge 0\}$$

(a) Solving (ILP):

$$\min_{x\in G} c^T \quad \text{with } G := Q \cap \mathbb{Z}^n, \quad Q := \{x \mid Ax \le b\}$$



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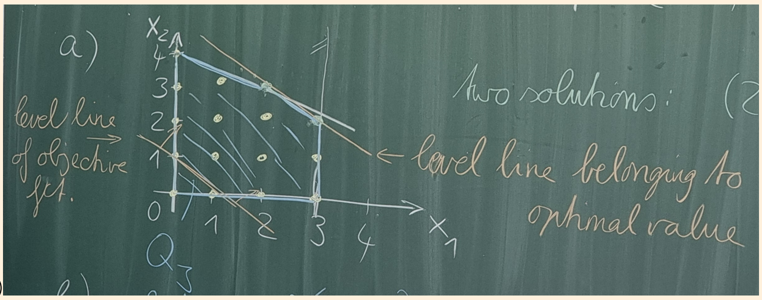
(b) We consider CILP

$$\min_{x \in G_I := \operatorname{conv}(G)} c^T x$$

Solution of (CILP): any point on the connection line between $(2, 3)$ and $(3, 2)$



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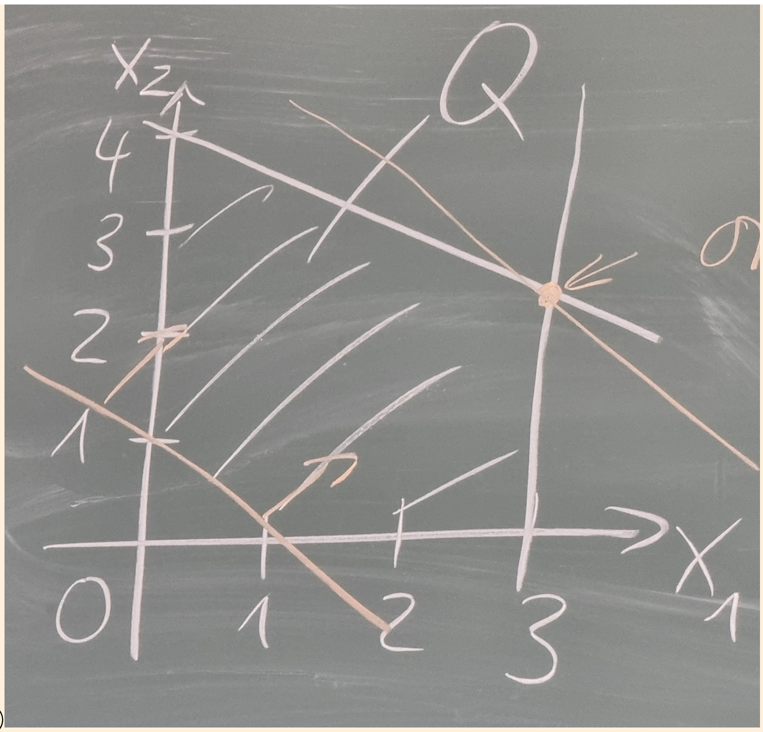
(c) We consider (CR):

$$\min_{x \in Q} c^T x$$



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Is Q an integer polyhedron? No, because it has a vertex whose components are not all integer.

$Q = \{x \in \mathbb{R}^n_+ \mid Ax \leq b\}$ integer $\iff$ all vertices integer $\iff$ $Q = Q_I$ $\Longleftarrow$ A is totally unimodular.

Exercise 3.

(a)

$$P_1 = \{x \in \mathbb{R}^3_+ \mid -x_1 + x_2 - x_3 \leq 5,\ x_1 + x_3 \leq 4\} = \{x \in \mathbb{R}^3_+ \mid Ax \leq b\}$$

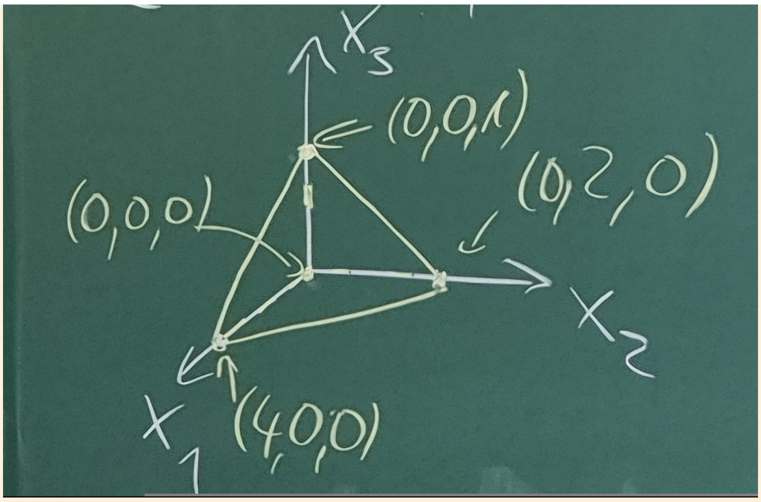
with $A = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \end{pmatrix}$, $b = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$ A is TU because:

- determinants of 1×1 submatrices are all $-1, 1$ or 0.
- Determinants of 2×2 submatrices are

$$\begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad \begin{vmatrix} -1 & -1 \\ 1 & 1 \end{vmatrix} = 0, \quad \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1$$

$\overset{2.22}{\Longrightarrow}$ P_1 is integer polyhedron.

$P_2 = \{x \in \mathbb{R}^3_+ \mid x_1 + 2x_2 + 4x_3 \leq 4\}$



All vertices are integer $\implies$ P_2 is integer polyhedron

(b) To justify: a linear program of the form

$$\min_{x \in P_2} c^T x$$

always has a solution.
This follows by Weierstrass Theorem (P_2 is nonempty, bounded and closed, $x \mapsto c^T x$ is continuous). The same argumentation can be used for (ILP)

$$\min_{x \in P_2 \cap \mathbb{Z}^3} c^T x$$

A.4 Tutorial 4

Exercise 1.

(a)

$$\max c^T x \quad \text{such that } a^T x \leq a_0,\ x \in [0,1]^n \tag{3}$$

Assumption: $\frac{c_i}{a_i} \geq \frac{c_{i+1}}{a_{i+1}}$

$$\widehat{x}_1 := ... = \widehat{x}_k := 1, \quad \widehat{x}_{k+1} := \frac{1}{a_{k+1}} \left(a_0 - \sum_{j=1}^{k} a_j \right), \widehat{x}_{k+2} := ... = \widehat{x}_n = 0$$

$$k := \max \left\{ j = 1, ..., n \mid \sum_{j=1}^{k} a_j \leq a_0 \right\}$$

To show: $\widehat{x}$ solves (3).

(i) show $\widehat{x}$ is feasible for (3): $x_i \in [0,1]$ for all $i = 1, ..., n$ by definition of $\widehat{x}$ and k.

$$\begin{aligned} a^T \widehat{x} &= \sum_{i=1}^{n} a_i \widehat{x}_i \\ &= \sum_{i=1}^{k} a_i + a_{k+1} \widehat{x}_{k+1} + 0 \\ &= \sum_{i=1}^{k} a_i + a_{k+1} \frac{1}{a_{k+1}} \left(a_o - \sum_{j=1}^{k} a_j \right) \\ &= a_0 \end{aligned}$$

(ii) Let x an arbitrary feasible point. To show: $c^T\widehat{x} \geq c^T x$.

$$\begin{aligned}
c^T\widehat{x} &= \sum_{i=1}^{k} c_i + c_{k+1}\frac{1}{a_{k+1}}\left(a_0 - \sum_{i=1}^{k} a_i\right) \\
&= \frac{c_{k+1}}{a_{k+1}}a_0 + \sum_{i=1}^{k}\left(c_i - \frac{c_{k+1}}{a_{k+1}}a_i\right)\cdot\frac{c_{k+1}}{a_{k+1}}a_0 \\
&\geq \frac{c_{k+1}}{a_{k+1}}\sum_{i=1}^{n} a_i x_i \\
&= \frac{c_{k+1}}{a_{k+1}}\sum_{i=1}^{k} a_i x_i + \underbrace{\sum_{k=k+1}^{n}\frac{c_{k+1}}{a_{k+1}}a_i x_i}_{\geq \sum_{i=k+1}^{n} c_i x_i}\cdot\sum_{i=1}^{k}\left(c_i - \frac{c_{k+1}}{a_{k+1}}a_i\right) \\
&\geq \sum_{i=1}^{k}\left(c_i - \frac{c_{k+1}}{a_{k+1}}a_i\right)x_i \\
&\implies c^T x^* \geq \frac{c_{k+1}}{a_{k+1}}\sum_{i=1}^{k} a_i x_i + \sum_{i=k+1}^{n} c_i x_i + \sum_{i=1}^{k}(c_i - \frac{c_{k+1}}{a_{k+1}}a_i)x_i \\
&= \sum_{i=1}^{n} c_i x_i = c^T x.
\end{aligned}$$

(b) S1: $\mathcal{G} = \{G\}$. $y = (1,0,0,1,1,1,0)$, $f(y) = 32 = \widehat{b}$, which is the lower bound for optimal objective function value (note: maximization problem!)
S2: Choose $X = G$. Divide it into

$$\begin{aligned}
X_1 &= \{x \in G \mid x_1 = 1\} \\
X_2 &= \{x \in G \mid x_1 = 0\}
\end{aligned}$$

S3: $b(X_1)$: solve continuous relaxation $\max_{x\in X_1} c^T x$.

$$\begin{aligned}
&\widehat{x} = \left(1, 1, \frac{7}{9}, 0, 0, 0, 0\right) \\
&\rightsquigarrow b(X_1) = c^T\widehat{x} = 36\frac{7}{9} \quad \text{upper bound for opt. obj. fct. on } X_1
\end{aligned}$$

$$\begin{aligned}
b(X_2) &: \widehat{x} = (0, 1, 1, \frac{1}{6}, 0, 0, 0) \\
&\rightsquigarrow c^T\widehat{x} = 35\frac{5}{6} \quad \text{upper bound on } X_2
\end{aligned}$$

S4:

- Choose $z \in X_1$, for example $z = (1,0,0,1,1,1,0)$. Again $f(z) = 32$
- Choose $z \in X_2$, f.e. $z = (0,1,1,0,0,0,0)$. $f(z) = 34$.

$\implies$ new lower bound for optimal objective function value: $\widehat{b} = 34$, $y = (0,1,1,0,0,0,0)$

Exercise 2.OPAL.

Exercise 3. Using B & B for the solution os TSP lower bounrds for optimal objective function value are needed. From the lecture the row/column reduction procedure is known.

$$C = \begin{pmatrix} \infty & 1 & 3 & 8 \\ 6 & \infty & 2 & 7 \\ 9 & 13 & \infty & 4 \\ 11 & 1 & 7 & \infty \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 4 \\ 1 \end{matrix}$$

$\alpha_i := \min\{c_{ij} \mid j = 1, ..., n\}$ and $\beta_j := \min\{c_{ij} - \alpha_i \mid i = 1, ..., n\}$. Here, the β_j are $4, 0, 0, 0$. Take $\gamma(C) := \sum_{i=1}^n \alpha_i + \sum_{j=1}^n \beta_j$ as lower bound for optimal function value. Alternative method: solving corresponding assignment problem

$$\delta(C) := \text{optimal obj. fct. value of this problem}$$

To show: $\delta(C) \geq \gamma(C)$. Assignement problem: $\max c^T x$ such that $Ax - b,\ x \geq 0$ with

$$\begin{aligned} x &= (x_{11}, ..., x_{1n}, ..., x_{n1}, ..., x_{nn})^T & &\in \mathbb{R}^{n^2} \\ c &= (c_{11}, ..., c_{1n}, ..., c_{n1}, ..., c_{nn})^T & &\in \mathbb{R}^{n^2} \\ b &= (1, ..., 1, 1, ..., 1)^T & &\in \mathbb{R}^{n^2} \\ A &= \text{picture 27.05 17h54} & &\in \mathbb{R}^{2n \times n^2} \end{aligned}$$

Dual problem:

$$\max b^T u \quad \text{such that } A^T u \leq c \text{ with } u \in \mathbb{R}^{2n}$$

Divide u by $u = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{R}^{n \times n}$.

$\implies$ dual problem:

$$\sum_{i=1}^n \alpha_i + \sum_{j=1}^n \beta_j \to \max \quad \text{such that } \alpha_i + \beta_j \leq c_{ij} \quad \forall i, j$$

(α, β) from row/column reduction procedure feasible for that problem.

$\implies \gamma(C) \leq$ optimal value of dual problem $= \delta(C)$.

A.5 Tutorial 5

Exercise 1.

nodes	a	v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8	v_9	b
$d(v)$	0	$\cancel{+\infty}$ 7	$\cancel{+\infty}$ 8	$\cancel{+\infty}$ 1	$\cancel{+\infty}$ 7	$\cancel{+\infty}$ 3	$\cancel{+\infty}$ 11	$\cancel{+\infty}$ 7	$\cancel{+\infty}$ $\cancel{8}$ 7	$\cancel{+\infty}$ 4	$\cancel{+\infty}$ $\cancel{10}$ 8
$p(v)$	a	$\cancel{v_1}$ a	$\cancel{v_2}$ a	$\cancel{v_3}$ a	$\cancel{v_4}$ v_3	$\cancel{v_5}$ v_3	$\cancel{v_6}$ v_1	$\cancel{v_7}$ v_5	$\cancel{v_8}$ $\cancel{v_5}$ v_9	$\cancel{v_9}$ v_5	$\cancel{b}$ $\cancel{v_9}$ v_8

Exercise 2.

(a) Let $\text{dist}(f)$ be the actual shortest distance from a to f. To show: $\forall f \in F$: $d(f) = \text{dist}(f)$ at the moment when f is added to A.
Assume that there is some f for which this is not true. WLOG let f be the first one with this property. Let P be an actual shortest path from a to f ($\rightarrow$ distance of $P = \text{dist}(f)$

- h... first node of P which does not belong to A (maybe $h = f$)
- g... predecessor of h on P (maybe $g = a$.)

$\Longrightarrow$ $g \in A$, i.e. it has already been considered in Disjkstra's algorithm.
$\Longrightarrow$ egde (g, h) was also considered already. $\rightarrow$ at this moment, $d(h) := d(g) + w_{(g,h)} = \text{dist}(g) + w_{(g,h)} = \text{dist}(h)$ was set. (note that any part of the a shortest path is again shortest path).
$\Longrightarrow$ $d(h) = \text{dist}(h) \leq \text{dist}(f) < d(f)$
Contradiction, because then h would have been considered instead of f.

(b)
- S1, S4: no operations
- S2: $\mathcal{O}(n)$ operations (or definitions) $[n := |F|]$
- S3:
  (a) $\mathcal{O}(n)$ (note that $F^+(f)$ has at most $n-1$ nodes)
  (b) $\mathcal{O}(n)$ (note that B has at most $n-1$ nodes)
  (c) $\mathcal{O}(n)$ (set operations union/set difference)

  S3 must be performed at most $n-1$ times (and therefore $\mathcal{O}(n)$ times) because in any run of S3 exactly one node more is added to A.

$\Longrightarrow$ whole complexity $= \mathcal{O}(n^2)$

Exercise 3.

(a) aim: shortest path problem from a to b as ILP

$$c^T x \rightarrow \min \quad \text{s.t.} \quad Ax = r, \; x \in \{0,1\}^{|E|}$$

- variables:

$$x_e := \begin{cases} 1, & \text{if } e \text{ is used in the path} \\ 0, & \text{if } e \text{ is not used.} \end{cases}$$

- objective function

$$\sum_{e \in E} w_e \cdot x_e \rightarrow \min$$

- constraints:

$$\forall v \in F \setminus \{a,b\} : \sum_{e \in F^+(v)} x_e - \sum_{e \in F^-(v)} x_e = 0$$

$$\text{for } a\text{: } \sum_{e \in F^+(a)} x_e - \sum_{e \in F^-(a)} x_e = 1$$

$$\text{for } b\text{: } \sum_{e \in F^+(b)} x_e - \sum_{e \in F^-(b)} x_e = -1$$

In matrix-vector form: $Ax = r$, where $A = \mathrm{In}_G \in \{-1,0,1\}^{|F|\times|E|}$ (incidence matrix) $\to \mathrm{In}_G$ is always totally unimodular by Theorem 2.24.

(b) (a) $\to$ feasible set empty

(b) $\to$ objective function unbounded from below on feasible set (goes to $-\infty$).

Exercise 4.input graph drawn 01.07 16h50
$x \in \mathbb{R}^{|E|}$ flow $\iff$

- $0 \le x_e \le w_e$
- $\sum_{e \in E^-(f)} x_e = \sum_{e \in E^+(f)} x_e$ for all $f \in F$.

x^1 is a flow.
flow value: $\mathrm{val}(x^1) = x_{(s,q)} = x_{(v_6,v_1)} = 20$
residual graph $G(x^1) = (F, R(x^1))$
edge e_{12} from v_1 to v_2: $r(e_{12}) = w_{e_{12}} - x_{e_{12}} + x_{e_{21}} = 17 - 10 + 0 = 7 > 0$
$e_{13} : r(e_{13} = 10 - 10 + 0 = 0 \implies r(e_{13}) \notin R(x^1)$
$e_{21} : r(e_{21}) = w_{e_{21}} - x_{e_{21}} + x_{e_{12}} = 10 > 0$
$e_{31} : r(e_{31}) = w_{e_{21}} - x_{e_{31}} + x_{e_{13}} = 0 - 0 + 10 = 10 > 0$
input residual graph picture 1.7 17h07
x^2 not a flow, because in v_5 we have $\sum_{e \in E^+(v_5)} x_e = 12 \neq 13 = \sum_{e \in E^-(v_5)}$
input residual graph picture 1.7 17h17
Problem related to max-flow problem: min-cut problem.
Cut any set $V \subset F$ with $s \in V$, but $q \notin V$.
Capacity: $\mathrm{cap}(V) = \sum_{e \in E^-(V)} w_e$, e.g. $\mathrm{cap}(\{v_4, v_5, v_6\}) = 9 + 14 = 23$
min cut $\{v_4, v_6\} \to \mathrm{cap}(...) = 21 = $ max. flow value

A.6 Tutorial 6

Exercise 1.

Exercise 2.

(a)

$$\begin{aligned}
\operatorname{val}(x) &= x_{(s,q)} \\
&= x_{(s,q)} + \sum_{f\in V}\left(\sum_{e\in E^-(f)} x_e - \sum_{e\in E^+(f)} x_e\right) \\
&= x_{(s,q)} + \sum_{e\in E^-(V)} x_e - \sum_{e\in E^+(V)} x_e \\
&= \sum_{e\in E^-(V)} - \sum_{e\in E^+(V)\setminus(s,q)} x_e \\
&\le \sum_{e\in E^-(V)} w_e \\
&= \operatorname{cap}(V)
\end{aligned}$$

(b) (i) $\implies$ (ii): x maximal flow, V min. cut. To show:

$$\forall e \in E^-(V) : e \notin R(x)$$

By Theorem 3.5: $\operatorname{val}(x) = \operatorname{cap}(V) \implies \sum_{e\in E^-(V)} x_e - \sum_{e\in E^+(V)\setminus\{(s,q)\}} x_e = \sum_{e\in E^-(V)} w_e \implies x_e = \begin{cases} w_e, & \forall e \in E^-(V) \\ 0, & \forall e \in E^+(V) \setminus \{(s,q)\}. \end{cases}$

$\implies \forall e = (f,g) \in E^-(V) : r(e) = w_e - x_e + x_{(g,f)} = 0.$

(ii) $\implies$ (iii) $\implies$ (i) OPAL.

(c) proof: $\nexists$ path from q to s in residual graph $G(x) \implies x$ is already maximal flow.

Define cut $V :=$ set of all nodes to which no path exists from q in $G(x)$. (in fact, $q \notin V$ and $s \in V$ hold).

Consider edges $e = (f,g) \in E^-(V)$, then there is a path from q to f in $G(x)$, but no path from q to g in $G(x)$ by definition of V. $\implies e \notin R(x)$. Because (i) $\iff$ (ii) holds, we deduce that x is a maximal flow (and V a min. cut).

Exercise 3.integer capacities $\overset{Theorem\ 3.8}{\implies}$ $\exists$ maximal flow whose components are all integer. BUT input counter example picture 1.7 18h05

A.7 Tutorial 7

Exercise 1.

(a) $G = (F, E)$ acyclic. To show: has exactly $|F| - |E|$ connected components. Induction regaring $k := |E|$.

- $k = 0$: G consits of isolated nodes only. $\implies$ each is its own connected component. $\implies$ $|F| - |E| = |F|$ connected components
- Induction hypothesis: graphs with $|F|$ nodes and k edges have $|F| - k$ connected components.
- Induction step: $k \to k + 1$: G graph with $|F|$ nodes and $k + 1$ edges. G' arises by removing one edge $\implies$ G' has k edges and $|F| - k$ connected components by induction hypothesis. G' has 1 connected component more than G (because G is acyclic) $\implies$ G has $|F| - k - 1 = |F| - (k + 1)$ connected components.

(b) $G' = (F, E')$ subgraph of G $\iff$ connected and acyclic.
(i) $\implies$ (ii): G' spanning tree . To show: G' has no cycles (✓) and $|F| - 1$ edges. Part (a) $\implies$ G' has $|F| - |E'|$ connected components.
On the other hand: G' connected $\implies$ $|F| - |E'| = 1$ $\implies$ $|E'| = |F| - 1$.
(ii) $\implies$ (iii): $G' = (F, E')$ has no cycles and $|F| - 1$ edges. To show: G' is connected (✓ because of part (a)) and has $|F| - 1$ edges (✓).
(iii) $\implies$ (i): $G' = (F, E')$ is connected and has $|F| - 1$ edges. To show: G' is spanning tree, i.e. connected + acyclic. Any connected graph must have at least $|F| - 1$ edges. If G' would contain a cycle, then there would not be enough edges to be connected.

Exercise 2.
We consider the problem

$$\min_{M \in \mathcal{M}} c(M) \tag{1}$$

from the lecture. Here $N := E$, $\mathcal{M} \subseteq \mathcal{P}(N)$ and $c(M) := \sum_{e \in M} c\{e\}$

(a) (F, M) minimum spanning tree. To show: M solves (1).

- M feasible, i.e. $M \in \mathcal{M}$? (F, M) has no cycles, because it is spanning tree. $\implies$ $M \in \mathcal{M}$ by definition.
- Assume there is $M' \in \mathcal{M}$ with $c(M') < c(M)$. WLOG, let (F, M') be connected and therefore spanning tree. (otherwise add edges until it is connected and still acyclic)

$$\begin{aligned}
&\implies |M'| = |M| = |F| - 1\\
&\implies c(M') = \sum_{e \in M'} c\{e\}\\
&= \sum_{e \in M} (w_e(\max\{w_a \mid a \in E\} + 1)\\
&= \sum_{e \in M'} w_e - (|F| - 1)(\max\{...\} + 1) = c(M)\\
&\implies \sum_{e \in M'} w_e < \sum_{e \in M} w_e
\end{aligned}$$

contradiction because (F, M) is minimum spanning tree.

(b) To show: (F, M) is a matroid.

- (M1): $\emptyset \in \mathcal{M}$ ✓
- (M2): Let $L, M \subseteq E$ with $m \in \mathcal{M}$ and $L \subset M$ be given. To show: $L \in \mathcal{M}$. $M \in \mathcal{M} \implies M$ has no cycles $\implies L \subseteq M$ has in particular no cycles because removing edges cannot lead to new cycles.
- (M3): Let $L, M \in \mathcal{M}$ with $|M| = |L| + 1$ be given. To show: $\exists e \in M \setminus L$: $L \cup \{e\}$ does not have cycles. (F, L) has exactly one connected component more compared to (F, M). $\implies$ there must be an edge $e \in M$ which connectes two different connected components of (F, L).
  $\implies$ $(F, L \cup \{e\})$ still acyclic and therefore $L \cup \{e\} \in \mathcal{M}$.

Exercise 3.

4b) $\max c^T x$ such that $5x_1 + 3x_2 + x_3 + x_4 \leq 8$, $x_i \in \{0, 1\}$.
$\to \min c(M)$ such that $M \in \mathcal{M} := \{M \subseteq \{1, 2, 3, 4\} : \sum_{i \in M} a_i \leq a_0\}$ $(N, \mathcal{M})$ violates (M3). for example $M = \{2, 3, 4\} \in \mathcal{M}$, $L = \{1, 2\} \in \mathcal{M}$. $\implies L \cup \{a\} \notin \mathcal{M} \quad \forall a \in M \setminus L$.

Greedy algorithm for $c = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$: $x_1 = 1 = x_2,\ x_3 = x_4 = 0$, but optimal solution $x_1 = 0$, $x_2 = x_3 = x_4 = 1$.